Uncertainty-Aware Fire Risk Assessment for UK Residential Buildings Using Public Data and Mechanism-Informed Simulation

Matthew Reid

July 13, 2026

Abstract

Fire risk assessment in the UK built environment remains largely periodic, qualitative, and compliance-oriented, despite the availability of public incident, building, demographic, and regulatory data. This paper develops an uncertainty-aware framework for residential fire-risk assessment and targeting in England. We combine public fire incident statistics, Energy Performance Certificate records, deprivation indices, Census occupancy proxies, care-home registers, and fire-service geography into an open, validated, UPRN-keyed (dwelling-level) database, and estimate calibrated posterior distributions for dwelling-fire occurrence and consequence-relevant outcomes. A hierarchical negative-binomial model predicts annual small-area dwelling-fire counts with held-out probabilistic calibration; record-level consequence models estimate serious spread and casualty associations under explicit treatment of alarm-state confounding, separating predictive association from intervention effect. To represent building mechanisms not observed in public registers, we introduce a transparent, mechanism-informed archetype fire/egress simulator, benchmark it against observed spread and casualty margins and a documented real-building boundary case, and compress its outputs into interpretable consequence surrogates. We then demonstrate decision-theoretic prevention targeting: under stated assumptions, a fixed smoke-alarm visit budget averts substantially more harm when allocated by posterior risk and expected benefit than by random or deprivation-only rules. The paper contributes a calibrated, auditable public-data decision framework for fire-risk prioritisation, and sets out a latent-state extension in which inspection findings are treated as noisy observations for live risk updating — providing a first partial instantiation on 1,441 published fire risk assessments: a fitted measurement model, a worked single-building posterior update, and first empirical bounds on assessor detection sensitivity, with full structural identification awaiting repeated assessment waves.

1 Introduction

Fire in the UK residential built environment remains a large and, recently, a growing problem: fire and rescue services attended 40,350 building fires in England in the year ending December 2025, up 5.5% year-on-year, of which 26,298 were dwelling fires [Ministry of Housing, Communities and Local Government, 2026]. Yet the fire risk assessment (FRA) that governs those buildings is still periodic, qualitative, and compliance-driven — checklist judgements recorded under the Regulatory Reform (Fire Safety) Order [UK Parliament, 2005] and codified methodologies such as PAS 79 [British Standards Institution, 2020a,b]. The Grenfell Tower fire and the Phase 2 Inquiry reframed residential fire safety around evidence, accountability, and the active management of higher-risk buildings [Grenfell Tower Inquiry, 2024], and the regulatory response — the Fire Safety Act 2021 [UK Parliament, 2021] and the Building Safety Act 2022 [UK Parliament, 2022], with their safety-case regime — demands exactly the kind of quantitative, evidence-updated risk picture that current practice does not produce.

Three gaps stand out. Risk is communicated as a categorical score rather than a distribution, despite evidence that risk matrices have poor resolution and can rank worse than random [Cox, 2008, Thomas et al., 2014]. Inspection findings are treated as ground truth rather than as noisy, intermittent observations; a targeted search across the fire-risk-assessment, inter-rater-reliability, assessor-variability, and PAS 79 literatures surfaced competence standards and professional guidance, but we are not aware of any empirical study estimating inter-assessor variability from repeated assessments of the same UK residential building — the sector's response has been a competence standard [British Standards Institution, 2025] rather than measurement of assessor noise — motivating the noisy-observation treatment we import from structural risk-based inspection [Straub and Faber, 2005, Luque and Straub, 2019]. And fire outcomes are rare events, yet calibration and uncertainty quantification (UQ) are seldom made explicit in operational FRA, even as reviews of fire machine learning confirm that UQ and calibration remain underdeveloped [Tapeh and Naser, 2022] against an established calibration-and-sharpness standard [Gneiting et al., 2007].

We recast FRA as Bayesian inference under uncertainty: a building's fire risk is a posterior distribution over ignition, spread, evacuation failure, and consequence severity, inferred from heterogeneous public evidence and updateable as new observations arrive. The novelty is not a new Bayesian network, a new fire simulator, or a new evacuation model in isolation; it is an auditable decision framework that couples calibrated UK open-data rare-event modelling, mechanistically grounded consequence simulation, uncertainty propagation, and decision-theoretic prevention targeting into one posterior-risk workflow. The paper is explicit throughout about what is built and what is specified: the calibrated public-data and simulation components below are *implemented*, while the latent-state, noisy-inspection extension for live risk updating (Section 3) is *proposed* — fully specified but not yet fully instantiated. Five contributions follow:

1. A UK open-data fire-risk database linking incident, building-stock, deprivation, occupancy, and vulnerability evidence at dwelling and small-area scale, validated against published totals (Section 4).
2. A calibrated rare-event occurrence model for dwelling fires — hierarchical negative-binomial with out-of-time probabilistic validation (Section 6).
3. Consequence models that explicitly separate predictive association from intervention effect, with formal treatment of alarm-state confounding, alongside a mechanism-informed archetype fire/evacuation simulator that supplies consequence estimates where public geometry and evacuation data do not exist (Section 5).
4. A decision-theoretic targeting demonstration: under stated assumptions, a fixed prevention budget allocated by posterior risk and expected benefit returns roughly nine times the value of untargeted allocation and 2.5 times deprivation-only targeting (Section 6.6).
5. A proposed latent-state formulation in which inspection findings enter as noisy observations (Section 3), motivated by the measurement-error tradition of structural risk-based inspection, together with a real-building stress test bounding the simulator's regime assumptions against Grenfell Tower (Section 6.7).

2 Related Work

We organise prior work along the eight themes that emerged from a systematic sweep of the literature, and position our integration against each. Seven differentiators recur: (A) UK FRA

/ Building-Safety-Act integration; (B) Bayesian latent-state inference of building condition; (C) calibrated uncertainty with missing-evidence flags; (D) inspection as noisy observation; (E) rare-event outcome treatment; (F) a mechanism-informed consequence surrogate; and (G) decision-theoretic inspection prioritisation.

UK fire-safety landscape and FRA methodology. A recent systematic review of UK building fire-safety practice catalogues its technological, mitigation, behavioural, and regulatory strands, and flags persistent compliance, enforcement, and maintenance gaps [Dauda et al., 2025]; the Hackitt review diagnosed a "broken" regulatory system and seeded the outcomes-based safety-case regime that followed [Hackitt, 2018, Perry, 2025]. Quantitative fire risk assessment itself has a long lineage — event-tree and deterministic consequence analysis in textbook form [Yung, 2008, Rasbash et al., 2004], statistics-driven residential risk indices [Xin and Huang, 2013], single-factor effectiveness estimates such as smoke-alarm impact [Gilbert and Butry, 2021], and UK-specific trend analyses of accidental dwelling fires [Taylor et al., 2026]. This literature motivates our setting (A) but is either policy diagnosis or static, deterministic assessment; what an "evidence-based" regime lacks, and what we supply, is the probabilistic latent-state machinery (B, C).

Probabilistic and Bayesian-network fire-risk models. The closest prior line chains fire development, escape, and intervention for UK dwellings in three-part Bayesian networks (BNs) [Matellini et al., 2018, 2013], alongside BN fatality and spread models [Hanea and Ale, 2009, Cheng and Hadjisophocleous, 2009] and a credal-network probabilistic risk assessment of fire occurrence applied to Grenfell Tower [Estrada-Lugo et al., 2019]. More recent variants extend the construction to heritage buildings [Chen et al., 2023], subway and tunnel scenario evolution [Li et al., 2024, Hidayat et al., 2025], and high-rise electrical and fuzzy composite scoring [Su et al., 2023, Wang et al., 2021a, Tan et al., 2021], all on the BN foundations of Pearl [1988]. We adopt this tradition's latent-state, posterior view of risk — but throughout it, conditional probability tables (CPTs) are fixed from expert judgement or aggregate statistics, inputs are treated as certain, and outputs are points: no calibrated uncertainty (C), no treatment of inspections as noisy observations (D), no physics-based consequence layer (F), and no inspection-prioritisation layer (G).

Uncertainty quantification and decision-theoretic inspection. Risk matrices — the format operational FRA actually uses — have poor resolution and can rank risks worse than random [Cox, 2008, Thomas et al., 2014], while calibration and sharpness under proper scoring rules provide the evaluation standard for probabilistic forecasts that we adopt [Gneiting et al., 2007]. Separately, the structural-reliability risk-based-inspection and value-of-information tradition treats inspections as imperfect information that updates a reliability estimate, and optimises inspection planning on that basis [Straub and Faber, 2005, Luque and Straub, 2019, Zhang et al., 2021, Vereecken et al., 2020, de Klerk et al., 2019]. Our contribution here is a transplant: importing inspection-as-noisy-observation (D) and expected value of inspection (G) into fire risk assessment, and replacing categorical scores with posterior distributions (C) — a combination this tradition has never instantiated for fire.

Fire-risk datasets, geospatial mapping, and service targeting. England-wide small-area evidence links deprivation to fire-related dwelling casualties [Li et al., 2022]; emergency-service demand analytics [Li et al., 2025] and evaluations of home-fire-safety-visit targeting [Waring et al., 2024] supply the operational context, with hierarchical Bayesian small-area rate estimation as the methodological template [MacNab, 2003]. The open-datacube precedent is IberFire, a spatio-temporal wildfire-risk cube assembled entirely from open sources [Erzibengoa et al., 2025]. These works supply our covariates and the open-data blueprint (A), but remain ecological and associational: none couples the data to a building-conditioned probabilistic model with quantified uncertainty (B, C).

Machine learning in fire safety. Reviews of fire-safety machine learning map a fast-growing field and flag precisely the weaknesses we target: underdeveloped uncertainty quantification and calibration [Yıldız, 2025, Tapeh and Naser, 2022], with mechanistically grounded learning advocated as the remedy [Naser, 2021]. Applications span smart-building design [Zeng and Huang, 2024], classifiers explained post hoc with SHapley additive explanations (SHAP) [Lu et al., 2023, Dai et al., 2025], and knowledge-graph and city-scale forecasting [Xu et al., 2025, Lee et al., 2025]. We use machine learning differently: as a calibrated surrogate *inside* a probabilistic model (C, F), rather than as a black-box or post-hoc-explained point classifier.

Fast fire-simulation surrogates. A systematic review of physics-informed surrogates finds them rarely embedded inside a probabilistic risk model [Yarmohammadian et al., 2025] — the integration gap this paper occupies. The component technology is mature: emulators predict fire fields, flashover, and detection orders of magnitude faster than computational fluid dynamics [Lattimer et al., 2020, Wang et al., 2021b, Nguyen et al., 2022, Zeng et al., 2022, 2024]; physics-informed neural networks learn interpretable spread parameters [Vogiatzoglou et al., 2025]; probabilistic diffusion surrogates output spread ensembles [Yu et al., 2026]; and graph neural networks serve as surrogates over building-block graphs [Wu et al., 2025]. We adopt the surrogate-on-graph idea for our archetypes (F) but couple it into the risk posterior with propagated uncertainty (C).

Coupled fire–evacuation modelling. The deterministic available-versus-required safe-egress-time (ASET/RSET) margin has been criticised as conceptually flawed [Babrauskas et al., 2010] and reframed distributionally [Schröder et al., 2020]; one-way fire-to-egress coupling and agent-based evacuation models degrade movement under tenability loss [Cheong et al., 2021, Wang et al., 2025], with occupant vulnerability shaping evacuation outcomes [Tancogne-Dejean and Laclémence, 2016]. We follow the distributional turn to its conclusion: evacuation failure is a node inferred *within* the risk model with propagated uncertainty (C, E), not a deterministic single-scenario margin.

Rare-event statistics. Extreme-value and peaks-over-threshold methods model heavy-tailed fire losses [Ramachandran, 1974, McNeil, 1997, Embrechts et al., 1997], and survival models give an interpretable time-to-event template [Kalair and Connaughton, 2021]. We apply this machinery to rare-event fire outcomes coupled to a building-conditioned posterior (E) — a link absent from its actuarial and traffic source domains.

Positioning. Our closest competitors, in order: the UK dwelling-fire Bayesian-network line [Matellini et al., 2018, 2013], which shares our setting and graphical structure but fixes its CPTs, treats inputs as certain, and offers neither calibrated intervals nor a latent-state posterior, inspection-noise model, physics surrogate, or prioritisation layer; the structural risk-based-inspection tradition [Straub and Faber, 2005, Luque and Straub, 2019], the closest analogue to our contributions (D) and (G) but developed for structural deterioration and never instantiated for fire; the physics-informed surrogate line [Yarmohammadian et al., 2025, Naser, 2021], which defines the surrogate half of (F) but almost never embeds it in a risk model with quantified uncertainty; and the risk-matrix critique

[Cox, 2008, Thomas et al., 2014], which justifies abandoning categorical outputs (C) but supplies no constructive replacement. Every competitor covers one or two of the seven differentiators; none couples UK regulatory context, a Bayesian latent-state model of building condition, calibrated uncertainty with missing-evidence flags, inspection-as-noisy-observation, rare-event outcome treatment, a mechanistic consequence surrogate, and decision-theoretic inspection prioritisation in a single framework. That integration is the novelty claim, and Section 6 tests its components rather than asserting them.

3 Probabilistic Model of Building Fire Risk

We treat the fire risk of a residential building b not as a categorical score but as a *posterior distribution* over the coupled processes of ignition, fire development, evacuation failure, and consequence severity, updated as evidence arrives. This section formalises that view.

3.1 Risk as expected loss

Let $I_b(t)$ denote the event that a fire ignites in building b during year t, and let $C_b(t)$ denote the resulting consequence (loss). We define the annualised risk as the expected loss,

$$R_b(t) \;=\; P\big(I_b(t)\big) \,\cdot\, \mathbb{E}\big[\, C_b(t) \,\big|\, I_b(t) \,\big], \tag{1}$$

i.e. the product of ignition probability and expected consequence given ignition. This mirrors the decomposition used in probabilistic safety analysis and in Bayesian-network dwelling-fire models [Matellini et al., 2018, Estrada-Lugo et al., 2019].

3.2 Consequence decomposition

Consequence is not a single quantity: a fire can destroy property without harming anyone, and can injure without spreading beyond the room of origin. We therefore decompose expected consequence as a *sum over loss channels*,

$$\mathbb{E}[C_b \mid I_b] \;=\; \mathbb{E}[C_b^{\text{prop}} \mid I_b] + \mathbb{E}[C_b^{\text{inj}} \mid I_b] + \mathbb{E}[C_b^{\text{fat}} \mid I_b], \tag{2}$$

and, within each channel, marginalise over the fire-development pathway rather than forcing every loss through a single chain. Writing S for the extent of spread (an ordered variable: contained to item, room of origin, beyond room, beyond floor) and E_f for evacuation failure,

$$P(\text{casualty} \mid I_b) \;=\; \sum_{s} \sum_{e\in\{E_f,\bar{E}_f\}} P(\text{casualty} \mid e, s, I_b)\, P(e \mid s, I_b)\, P(s \mid I_b), \tag{3}$$

which preserves the causal ordering ignition $\to$ spread $\to$ egress $\to$ harm while permitting casualties without major spread (e.g. smoke inhalation or fire-fighting attempts in the room of origin, the $s{=}$room, $e{=}\bar{E}_f$ terms) and property loss without evacuation failure. The dominant term for fatalities is the one highlighted by the multiplicative chain $P(S{\geq}\text{room} \mid I)\, P(E_f \mid S, I)\, \mathbb{E}[L \mid E_f, S, I]$, which we retain as the leading-order approximation used in the worked examples; Equation (3) is the full object. Each factor depends on the building's physical and occupancy characteristics; the egress factors are where the physics-based consequence surrogate of Section 5 enters, and the property channel is valued directly from incident-damage statistics.

Why this factorisation. Four modelling choices in Equation (3) warrant defence. First, spread is marginalised *before* egress because the two orderings coincide: compartmentation failure is physically upstream of tenability loss, so conditioning harm on spread respects the causal chain, and spread is also the margin the national incident record actually reports, so the factorisation matches an observable quantity rather than a latent one. Second, evacuation is kept binary (E_f versus $\bar{E}_f$) because the record carries no evacuation *timing*, only its outcome; a finer egress state would be prior, not data. Third, suppression is not a separate factor: sprinkler coverage in the existing English dwelling stock is of the order of one per cent, so it is negligible as a population term, and brigade intervention acts *after* the spread outcome that defines our margin, so its effect is already absorbed into the empirically estimated $P(s \mid I_b)$ rather than double-counted. Fourth, we do not carry a fully joint latent $(\text{spread}, \text{occupancy}, \text{detection})$ object here: that model is written down in Section 3.4, but it is not identified from a single cross-section, so the factorised form of Equation (3) is the part the present data support.

3.3 Latent building state and evidence

The factors in Equations (1)–(3) are governed by a *latent building state*

$$z_b \;=\; \{\,\text{compartmentation},\ \text{fuel load},\ \text{alarm reliability},\ \text{suppression},\\ \text{occupancy vulnerability},\ \text{maintenance quality},\ \text{electrical risk}\,\}, \tag{4}$$

which is never observed directly. Instead we observe *evidence*

$$x_b \;=\; \{\,\text{inspection findings},\ \text{public building attributes},\\ \text{incident history},\ \text{occupancy proxies},\ \text{local covariates}\,\}. \tag{5}$$

Inference proceeds in two steps: recover the posterior over latent state,

$$P(z_b \mid x_b) \;\propto\; P(x_b \mid z_b)\, P(z_b), \tag{6}$$

and propagate it through the risk model to obtain the posterior risk distribution,

$$P(R_b \mid x_b) \;=\; \int P(R_b \mid z_b)\, P(z_b \mid x_b)\, \mathrm{d}z_b. \tag{7}$$

Equation (7) is the central object of the framework: *risk is a distribution, not a score.* Its spread quantifies uncertainty arising from missing or unreliable evidence, and its credible intervals and missing-evidence flags are the operational outputs consumed by FRA workflows (Section 6).

Implemented vs proposed. This paper implements calibrated components of Equations (1)–(3) — a small-area occurrence model for $P(I)$, record-level consequence models, and the consequence simulator — composed with propagated uncertainty in Sections 6.5 and 6.6. The joint latent-state posterior of Equations (6)–(7), with the variables of Equation (4), is specified in full but only partially instantiated: Section 6.8 fits its measurement model on 1,441 published fire risk assessments, while the transition dynamics remain unidentified until repeated assessment waves exist.

Predictive association vs intervention effect. Throughout we distinguish the *predictive* quantity $P(Y{=}1 \mid x, I)$, which record-level regressions estimate, from the *interventional* contrast $P\big(Y^{\mathrm{do(alarm=working)}}{=}1\big) - P\big(Y^{\mathrm{do(alarm=absent)}}{=}1\big)$. Observational fire data conflate the two: alarm operation, occupancy, and casualty are mutually selected, since an alarm operates, and a casualty can occur, only when someone is present. Predictive associations are therefore used for risk ranking, while intervention effects are taken from the mechanism-informed simulator, whose alarm effect is interventional by construction (Section 5). The distinction is drawn here, before any results, because Section 6 reports estimates whose observational sign differs from the interventional one.

3.4 Inspection as noisy observation

We treat an inspection not as ground truth but as a noisy, intermittent measurement of the latent state, in the spirit of detector/measurement-error modelling. For a latent attribute $z_b^{(k)}$ (e.g. compartmentation integrity), an inspection at time t yields an observation $o_b^{(k)}(t)$ through an observation model

$$o_b^{(k)}(t) \;\sim\; P\big(o \mid z_b^{(k)},\, \phi_k\big), \tag{8}$$

where ϕ_k captures inspection sensitivity, specificity, and inspector variability. Absence of a recent inspection simply means no update to $P(z_b^{(k)} \mid x_b)$, so uncertainty grows with time since last observation.

Latent-state dynamics, and what the data instantiate. Indexing the state by time, $z_{b,t}$ follows a first-order state space with interventions,

$$\begin{aligned}
z_{b,t} &\;\sim\; p\big(z_{b,t} \mid z_{b,t-1},\, u_{b,t}\big) && \text{(slow drift; jumps at refurbishment/intervention } u_{b,t}\text{)}, && (9)\\
o_{b,t} &\;\sim\; p\big(o_{b,t} \mid z_{b,t},\, \phi\big) && \text{(inspection observations, Eq. (8))}, && (10)\\
y_{b,t} &\;\sim\; p\big(y_{b,t} \mid z_{b,t},\, x_{b,t}\big) && \text{(incident outcomes: the models of Section 6)}, && (11)
\end{aligned}$$

so a new inspection updates $P(z_{b,t} \mid o_{b,1:t}, y_{b,1:t}, x_{b,1:t})$ by standard filtering, and drift between inspections widens it. The assessor parameters ϕ — detection sensitivity, specificity, between-assessor variance — are the empirically hardest piece. Section 6.8 fits the measurement model of Equation (10) on 1,441 published fire risk assessments across 653 Camden estates and *bounds* detection sensitivity from their recurrence structure; point identification of ϕ and of the transition dynamics in Equation (9) needs repeated assessment waves of the same buildings. We specify the model now because the repeated assessments mandated under the Building Safety Act will identify these parameters; the contribution here is the framework and the first empirical persistence estimates, not a fitted latent state.

This framing also yields the *expected value of inspection* (EVI): with actions a and losses L, acquiring observation o_b before acting is worth

$$\mathrm{EVI}(b) \;=\; \min_a \mathbb{E}\big[L(a) \mid x_b\big] \;-\; \mathbb{E}_{o_b}\Big[\min_a \mathbb{E}\big[L(a) \mid x_b, o_b\big]\Big] \;\geq\; 0, \tag{12}$$

the classic value-of-information quantity: it is largest where the posterior is both consequential and undecided, and zero where the optimal action is already determined. Section 6.6 instantiates Equation (12) for alarm-ownership inspection and shows the resulting ranking differs materially from expected-risk ranking.

3.5 Decision layer: allocating a prevention budget

The posterior risk distributions exist to drive decisions, so the framework includes the allocation problem explicitly. Let area i contain H_i dwellings, let n_i be the number visited under a prevention programme with budget B, and let b_i be the expected present-value averted loss per visited dwelling — composed from the posterior fire rate, the probability the dwelling lacks a working alarm, and the interventional consequence reduction of Section 5. The allocation problem is

$$\max_{n} \sum_i n_i\, b_i \qquad \text{s.t.} \quad \sum_i n_i \le B, \quad 0 \le n_i \le H_i. \tag{13}$$

Because every dwelling within an area shares the same marginal value $b_i \ge 0$, Equation (13) is a continuous knapsack with box constraints, and the greedy rule the implementation uses — fill areas in decreasing order of b_i until the budget is exhausted — is exactly optimal.

Proposition 1 (Greedy optimality of the allocation)**.** *Let $b_i \ge 0$ and index the areas so that $b_{(1)} \ge b_{(2)} \ge \cdots$. In the continuous relaxation of Equation (13) ($n_i \in [0, H_i]$, $\sum_i n_i \le B$), the greedy allocation that fills areas in this order — $n_{(1)} = H_{(1)}, n_{(2)} = H_{(2)}, \ldots$ until the running total reaches B, filling the final boundary area only partially and setting the remainder to zero — attains the maximum of $\sum_i n_i b_i$.*

Proof. A maximiser exists: the objective is linear and the feasible set compact. Let n^* be any maximiser and suppose it is not of the greedy form. Then there is a pair (j, k) with $b_j > b_k$, $n^*_j < H_j$ and $n^*_k > 0$: a higher-value area not yet full while a lower-value one still draws budget. Moving $\delta = \min(H_j - n^*_j,\ n^*_k) > 0$ from k to j leaves $\sum_i n_i$ unchanged and both box constraints satisfied, so the result is feasible, and it raises the objective by $\delta\,(b_j - b_k) > 0$ — contradicting the optimality of n^*. Every maximiser is therefore of the greedy form, and all greedy-form allocations share the same objective value (ties $b_j = b_k$ only permute budget between equal-value areas), so the greedy allocation attains the maximum. (If $B \ge \sum_i H_i$ the budget binds nowhere and $n_i = H_i$ is optimal because $b_i \ge 0$.) □

Remarks. Were the per-area benefit strictly concave, $g_i(n_i)$ with g_i' non-increasing, the KKT conditions of the same concave programme would replace "fill in order" by *water-filling*: interior areas equalise their marginal benefit $g_i'(n_i) = \mu$ at the budget multiplier μ, and the linear case is the degenerate limit in which a constant marginal $g_i' = b_i$ forces each area to fill completely before the next opens. Because visits are integer dwellings, the deployed allocation rounds only the single boundary area, so the gap between the integer optimum and the continuous one is at most the benefit of one dwelling there, $\le \max_i b_i$ — negligible against a budget of order 10^5 visits.

Uncertainty propagates by solving Equation (13) per posterior draw, giving credible intervals on the value of any policy; and replacing b_i with EVI_i of Equation (12) yields the inspection-prioritisation variant.

Propagating posterior uncertainty to the decision. The pipeline is a posterior $\pi(\theta \mid y)$, a loss $L(a, \theta)$ evaluated through the simulator, and a plug-in decision that minimises the posterior expected loss. Two guarantees govern how uncertainty in θ reaches the chosen action.

Assumption 1 (Lipschitz loss)**.** *The action set A is finite, and for each $a \in A$ the loss $L(a, \cdot)$ is L_a-Lipschitz in θ on a set containing both the support of $\pi(\theta \mid y)$ and the point estimate $\hat{\theta}$, i.e. $|L(a, \theta) - L(a, \theta')| \le L_a \|\theta - \theta'\|$. Write $L_{\max} = \max_{a \in A} L_a$.*

Proposition 2 (Propagation and decision robustness)**.** *Write $\bar{L}(a) = \mathbb{E}_\pi[L(a, \theta) \mid y]$ for the posterior expected loss and $a^\star = \arg\min_a \bar{L}(a)$ for the Bayes action.*

(i) (Monte Carlo propagation.) *For S posterior draws $\theta_s \sim \pi(\cdot \mid y)$, the estimator $\hat{L}_S(a) = S^{-1} \sum_{s=1}^{S} L(a, \theta_s)$ is unbiased for $\bar{L}(a)$ with standard error $\sqrt{\mathrm{Var}_\pi[L(a, \theta)] / S}$. If the draws*

are importance-weighted with self-normalised weights w_s ($\sum_s w_s = 1$), the estimator $\sum_s w_s\, L(a, \theta_s)$ has, by the delta method, variance approximately $\mathrm{Var}_\pi[L(a,\theta)]\,/\,S_{\text{eff}}$ with effective sample size $S_{\text{eff}} = 1/\sum_s w_s^2$ — the importance-sampling effective sample size of Kong (1992).

(ii) (Decision robustness.) *Let $\hat\theta$ be any point estimate and $\hat a = \arg\min_a L(a, \hat\theta)$ the plug-in action. Under Assumption 1 the Bayes regret of acting on $\hat\theta$ obeys*

$$\bar L(\hat a) - \bar L(a^\star) \;\le\; 2\, L_{\max}\, \mathbb{E}_\pi \|\theta - \hat\theta\|. \tag{14}$$

Proof. Part (i) is the standard Monte Carlo variance, with the self-normalised importance-sampling correction $S_{\text{eff}} = (\sum_s w_s)^2/\sum_s w_s^2 = 1/\sum_s w_s^2$ for normalised weights. For part (ii), write $\varepsilon = \mathbb{E}_\pi\|\theta - \hat\theta\|$. For every a,

$$|\bar L(a) - L(a,\hat\theta)| = \big|\,\mathbb{E}_\pi[L(a,\theta) - L(a,\hat\theta)]\,\big| \le \mathbb{E}_\pi\,|L(a,\theta) - L(a,\hat\theta)| \le L_a\,\varepsilon \le L_{\max}\,\varepsilon.$$

Decompose the regret by inserting $L(\cdot,\hat\theta)$:

$$\bar L(\hat a) - \bar L(a^\star) = \underbrace{[\,\bar L(\hat a) - L(\hat a,\hat\theta)\,]}_{\le L_{\max}\varepsilon} + \underbrace{[\,L(\hat a,\hat\theta) - L(a^\star,\hat\theta)\,]}_{\le 0} + \underbrace{[\,L(a^\star,\hat\theta) - \bar L(a^\star)\,]}_{\le L_{\max}\varepsilon}.$$

The middle bracket is non-positive because $\hat a$ minimises $L(\cdot,\hat\theta)$; the outer two are each at most $L_{\max}\varepsilon$ by the preceding display. Summing gives Equation (14). □

Part (ii) makes the value of sharp inference concrete: posterior contraction from full-data inference (Section 6) shrinks $\mathbb{E}_\pi\|\theta - \hat\theta\|$ and so tightens the regret bound directly, meaning a point-estimate decision forfeits least precisely where the posterior is already concentrated. Section 6.6 evaluates both allocation variants against random and deprivation-based allocation.

3.6 Rare events and calibration

Building-level fire is a rare event: annual ignition probabilities are small and severe outcomes rarer still [Ministry of Housing, Communities and Local Government, 2026]. This has two consequences for the model. First, estimation must borrow strength across buildings via hierarchical/partial-pooling priors on $P(I_b)$ and on the consequence factors, rather than fitting each building independently. Second, *calibration* is a first-class objective: predicted probabilities and credible intervals must match observed frequencies at the population level. We therefore evaluate the model with proper scoring rules and calibration diagnostics (reliability curves, coverage of credible intervals) against historical English incident statistics, treating the aggregate counts as the frequency ground truth for the rare-event process (Section 6).

3.7 From specification to fit: estimation and evaluation machinery

The results of Section 6 instantiate the framework with three fitted likelihoods; we state them fully here so that every fit is reproducible from the paper alone.

Occurrence. Dwelling-fire counts y_{it} in LSOA i and fiscal year t follow a hierarchical negative binomial in its mean–dispersion (NB2) form,

$$y_{it} \;\sim\; \mathrm{NB2}(\mu_{it},\,\phi), \qquad \mathrm{Var}[y_{it}] = \mu_{it} + \mu_{it}^2/\phi, \tag{15}$$

$$\log \mu_{it} \;=\; \log H_i \;+\; \alpha \;+\; \mathbf{x}_{it}^\top \boldsymbol\beta \;+\; b\,(t - \bar t) \;+\; u_{f(i)} \;+\; s_i, \tag{16}$$

with households H_i as exposure offset, ten standardised covariates $\mathbf{x}_{it}$, a linear year trend, service random intercepts $u_f \sim \mathcal{N}(0, \sigma_u^2)$, and — in the preferred M5 specification — a BYM2 spatial effect $s_i = \sigma_s(\sqrt{1-\rho}\, v_i + \sqrt{\rho}\, w_i/\sqrt{\kappa})$ combining unstructured noise $v_i \sim \mathcal{N}(0,1)$ with an intrinsic conditional autoregression $\mathbf{w} \sim \mathrm{ICAR}(\mathbf{A})$ on the queen-contiguity LSOA adjacency $\mathbf{A}$, scaled by the graph's marginal variance κ so that σ_s and the mixing weight ρ are interpretable. Priors are weakly informative: $\alpha \sim \mathcal{N}(-6.5, 1.5^2)$ centred on the historical national log-rate, $\beta_k \sim \mathcal{N}(0,1)$, $b \sim \mathcal{N}(0, 0.5^2)$, $\sigma_u \sim \mathrm{HalfNormal}(0.5)$, and $1/\phi \sim \mathrm{HalfNormal}(1)$, the last parameterised through the inverse concentration so the Poisson limit sits at the prior mode's edge rather than at a boundary singularity — the reparameterisation that restores tractable posterior geometry.

Consequence. Record-level outcomes (serious spread; any casualty) follow hierarchical logistic regressions,

$$y_{jf} \sim \mathrm{Bernoulli}(p_{jf}), \qquad \operatorname{logit} p_{jf} = \alpha + \mathbf{x}_j^\top \boldsymbol{\beta} + u_f, \qquad u_f \sim \mathcal{N}(0, \sigma_u^2), \tag{17}$$

with reference-coded factors for dwelling type, cause, ignition power, occupancy, time of day, the four-level alarm factor, a standardised year trend and the service-year response-time control; priors $\alpha \sim \mathcal{N}(0, 5^2)$, $\beta \sim \mathcal{N}(0, 2.5^2)$, $\sigma_u \sim \mathrm{HalfNormal}(1)$.

How the fits are performed. Exact posteriors are drawn with the No-U-Turn Sampler (NUTS), the adaptive Hamiltonian Monte Carlo variant that simulates Hamiltonian trajectories on the log-posterior and eliminates hand-tuned step counts. Hierarchical models of this type develop funnel-shaped geometry at full data scale on which the NUTS step size collapses; we therefore follow a two-track protocol. Track one: exact NUTS on stratified subsamples (∼10,000 LSOAs spanning every service; ∼30,000 fires stratified by fiscal year), 2–4 chains, 300–1,000 warmup and post-warmup draws, accepted only with $\hat{R} \leq 1.02$, bulk effective sample sizes above ∼150 for every reported quantity, and zero divergent transitions. Track two: the identical model on the *full* data by automatic-differentiation variational inference (ADVI), fitting a full-rank Gaussian $q_\lambda(\boldsymbol{\theta})$ by maximising the evidence lower bound

$$\mathrm{ELBO}(\lambda) \;=\; \mathbb{E}_{q_\lambda}\big[\log p(\mathbf{y}, \boldsymbol{\theta})\big] \;-\; \mathbb{E}_{q_\lambda}\big[\log q_\lambda(\boldsymbol{\theta})\big], \tag{18}$$

which serves as a point-estimate cross-check on all the data: coefficients must agree with the subsample NUTS posterior within its credible intervals. Known variational failure modes — ridge-splitting between an intercept and its random-effect mean, understated group-level predictive uncertainty — are checked for and reported wherever they occur.

Where exact NUTS is feasible on the full data, it supersedes both tracks: the consequence models (logistic likelihood, no dispersion parameter) converge cleanly on all 451,172 records and those posteriors are the ones reported. The occurrence model does not admit full-data NUTS: on all 202,530 training rows the negative-binomial posterior concentrates onto a ridge between the intercept, the service-effect mean and the dispersion parameter that defeats every standard reparameterisation — non-centred random effects mix at effective sample sizes of single digits, the centred form collapses the adapted step size to 4×10^{-6}, and a QR-orthogonalised design (which removes the deprivation-domain collinearity) still collapses to 5×10^{-7}. Sampling difficulty is however a *traversal* problem, and importance sampling requires only *coverage*: with 58 unconstrained parameters and 2×10^5 observations the posterior is close to Gaussian (Bernstein–von Mises), so the exact full-data posterior is recovered by Pareto-smoothed importance sampling (PSIS) [Vehtari et al., 2024] over a Laplace proposal. Concretely: the model is written in its *centred* parameterisation — hopeless for NUTS, but the right coordinates here, since each service's thousands of observations pin its effect directly and the residual intercept–service-mean ridge is linear, which a Gaussian proposal absorbs exactly through the Hessian; the mode and dense Hessian are computed in unconstrained space; a multivariate Student-t ($\nu = 8$, scale 1.1) with that mean and covariance proposes 50,000 draws; and the importance weights $w_s \propto p(\boldsymbol{\theta}_s \mid \mathbf{y})/q(\boldsymbol{\theta}_s)$ are stabilised by generalised-Pareto smoothing of the largest weights. The smoothing diagnostic certifies the result: $\hat{k} = 0.33$, far below the 0.7 reliability threshold, with an importance-sampling effective sample size of 16,892 — and the recovered posterior matches the subsample NUTS, ADVI and maximum-likelihood (MLE) fits on every parameter. The two-track protocol is thus retained as scaffolding and cross-check, while every headline posterior in this paper is an exact full-data object.

How calibration is measured. For discrete counts, central-interval coverage over-covers structurally, so the primary diagnostic is the randomised probability integral transform: for held-out count y with predictive cumulative distribution F,

$$u \;=\; F(y-1) \;+\; v\,[F(y) - F(y-1)], \qquad v \sim \text{Uniform}(0,1), \tag{19}$$

which is exactly $\text{Uniform}(0,1)$ if and only if the predictive distribution is calibrated; we report the Kolmogorov–Smirnov distance of u from uniformity and the randomised coverage of central 50%/90% intervals. Binary outcomes are scored with the Brier score $\frac{1}{n}\sum_j (p_j - y_j)^2$, the mean log score, and the expected calibration error $\text{ECE} = \sum_m \frac{n_m}{n}\,|\bar{p}_m - \bar{y}_m|$ over $M{=}10$ equal-width probability bins, alongside coverage of service-level posterior-predictive count intervals. All evaluation is out-of-time: the final one or two fiscal years are never seen in fitting.

Fitting the measurement model. The latent-state instantiation of Section 6.8 operationalises Equations (10)–(11) cross-sectionally: a scalar latent deficit $\theta_e \sim \mathcal{N}(0,1)$ per estate e loads on the assessment indicators — ordinal overall rating via a cumulative-logit link, priority-A and priority-B action counts via Poisson links $\log \mathbb{E}[\text{count}] = a_k + \lambda_k \theta_e$, and deficiency-theme flags via Bernoulli-logit links — while the structural outcome is the estate's long-run fire count,

$$y_e \sim \text{NB2}(\mu_e, \phi), \qquad \log \mu_e = \log E_e + \alpha + \beta_\theta\,\theta_e + \beta_{\text{IMD}}\,\text{IMD}_e, \tag{20}$$

with exposure E_e (flats $\times$ years) and orientation pinned by constraining the priority-A loading positive. The posterior update shown in Figure 10 is then literal conditioning: the block's $P(\theta_e \mid \text{FRA indicators})$ propagated through Equation (20) to a fire-rate posterior, before and after the indicators are observed.

4 Data

The pipeline assembles eight public English datasets — most published by the Ministry of Housing, Communities and Local Government (MHCLG) or the Office for National Statistics (ONS) — into a single analysis-ready database, all joined on the 2021-Census Lower-layer Super Output Area (LSOA; code `lsoa21`, 33,755 in England). Table 1 lists the sources. Every dataset is published under the Open Government Licence (OGL) v3.0, with the single exception noted below for address-level Energy Performance Certificate (EPC) records.

Sources. Fire incidence comes from the MHCLG fire statistics collection [Ministry of Housing, Communities and Local Government, 2026]: three record-level datasets and a suite of aggregate FIRE-series benchmark tables. The *dwelling-fires* dataset (one row per primary dwelling fire

Table 1: Public data sources for the pipeline. Grain is the native unit of observation; all are keyed or aggregated to the LSOA for modelling. Row counts are as ingested. MHCLG: Ministry of Housing, Communities and Local Government; ONS: Office for National Statistics; EPC: Energy Performance Certificate; IMD: Index of Multiple Deprivation; CQC: Care Quality Commission.

Dataset	Grain	Rows	Role in model
Dwelling fires (MHCLG)	incident	451,172	$P(\text{casualty})$, $P(S)$
Other-building fires (MHCLG)	incident	231,426	non-dwelling context
Low-level geography (MHCLG)	incident	8,502,911	$P(I)$ spatial exposure
EPC register (MHCLG)	certificate	22,375,762	building attributes (x_b)
— latest per dwelling	dwelling	17,302,650	dwelling-stock aggregates
IMD 2025 (MHCLG)	LSOA	33,755	deprivation covariates (x_b)
Census 2021 (ONS, 5 tables)	LSOA	33,755	occupancy proxies (x_b)
CQC active locations	premises	56,870	occupancy vulnerability
Postcode → LSOA lookup (ONS)	postcode	2,714,964	EPC join key

attended by a fire and rescue service (FRS), FY 2010/11–2024/25) carries the ignition, alarm, spread and casualty fields used by the consequence models; the *low-level-geography* dataset locates every incident to an LSOA and supplies the spatial exposure denominator. We ingested the full 92-table FIRE catalogue and retain 61 benchmark tables, including FRS response times (FIRE1001, used as a consequence covariate in Section 6.2), building-type spread and casualty rates (FIRE0301/0304, used to calibrate the institutional archetypes), and smoke-alarm operation rates (FIRE0701/0703/0705). Building attributes come from the EPC domestic register [Ministry of Housing, Communities and Local Government, 2025] — 22.38 M England lodgements, deduplicated to 17.30 M latest-per-dwelling certificates for stock aggregates (built form, construction-age band, floor area, main fuel, tenure, floor level). Local covariates come from the Index of Multiple Deprivation (IMD) 2025 (all seven domains per LSOA) and five Census 2021 tables [Office for National Statistics, 2022]: accommodation type (TS044), central heating (TS046), occupancy rating (TS052), tenure (TS054) and household composition (TS003). Occupancy vulnerability is sharpened by the Care Quality Commission (CQC) register of active health-and-social-care locations (14,896 flagged as care homes) and, as a worked example of the house-in-multiple-occupation (HMO) archetype, the London Borough of Camden HMO licensing register (no national open HMO dataset exists).

Join architecture. The canonical spatial key is the LSOA. IMD, Census and the EPC dwelling-stock aggregates join to it directly; MHCLG has re-coded the full incident history (including 2010/11) to 2021 LSOAs, so no 2011→2021 crosswalk is required. EPC certificates carry a postcode but no LSOA, so they are joined via the ONS best-fit UPRN join to the Ordnance Survey open identifier products, with postcode best-fit as fallback (99.91% of England lodgements matched, 98.6% exactly by UPRN). The record-level dwelling-fire dataset carries no geography below the FRS area (withheld for privacy), so those models are hierarchical over the 43–44 FRS areas rather than over LSOAs; the spatial occurrence model instead uses the LSOA-resolved incident counts against Census household denominators.

Validation. Twenty-seven automated checks pass (zero failures). Record-level dwelling-fire counts reconcile to the published FIRE0201 aggregate at a ratio of 1.000 in every year from 2010/11 to 2023/24 (residuals of a handful of incidents), and 0.995 in 2024/25 (a known Suffolk data gap, below). The 33,755-LSOA count matches exactly across IMD and all five Census tables; 99.36%

of low-level-geography rows carry a valid English LSOA that joins to IMD at 100% coverage. As an independent cross-check, the per-LSOA flat share derived from EPC agrees with the Census TS044 flat share at Pearson $r = 0.977$, confirming the postcode-to-LSOA join and the EPC stock aggregation.

Known quirks (carried honestly into the models). Four data-quality issues are documented and handled rather than hidden. (i) *Suffolk 2024/25 gap:* the dwelling dataset is missing Suffolk records for September 2024–March 2025 (212 fires short), so 2024/25 record-level counts sit just below the FIRE0201 aggregate and Suffolk is excluded from that year in the occurrence model; the principled treatment — marginalising the missing months under a partial-observation likelihood rather than dropping the area — is v2 work. (ii) *'Not known' LSOAs:* 54,194 incident rows carry an unresolved LSOA code (a publisher over-count for three services) and are flagged and excluded from spatial modelling via `is_valid_lsoa21`; treating the true LSOA as a latent allocation to be inferred, rather than discarding these rows, is likewise deferred to v2. (iii) *EPC `floor_level` is flat-only:* it is populated for flats/maisonettes (roughly 30% of stock) and blank for houses, so its 69% null rate is structural, not missing data, and it is used only after filtering to flats. (iv) *Address-level EPC provenance, by design:* the processed database holds *no* Royal Mail address strings — only the Unique Property Reference Number (UPRN), postcode, administrative geography and dwelling attributes. Dwelling-level linkage is via the ONS UPRN Directory (ONSUD; exact UPRN joins under OGL), with postcode best-fit as a fallback; the Royal Mail intellectual property that carries the address-level licence restriction exists only in the raw source, which we do not redistribute. Reproduction is therefore via the documented pipeline plus a free EPC-service registration, and every EPC quantity still enters the model as an LSOA-level aggregate.

5 Mechanism-Informed Consequence Surrogate

The final factors of Equation (3) — the probability of significant spread and of evacuation failure given ignition — depend on physical building geometry and occupant behaviour that no public register records directly. In place of real geometry, we represent each building as a synthetic *archetype graph*: nodes are typed rooms, corridors, stairs, lobbies and exits, and edges carry a compartmentation-barrier type (open, internal door, fire door, compartment wall) and a traversal length. Seven families cover the minimum-viable-product (MVP) scope — detached and terraced houses, low-rise and high-rise purpose-built flats, HMO, care-home-like wing, and mixed-use residential-over-commercial — parameterised by storeys, occupant count, mobility-impaired fraction, and number of exits.

Fire develops as a stochastic process on the graph. An ignition (located by room-kind weights derived from the dwelling-fires data, kitchens dominant) is most often minor and self-limits at the room of origin; a developing fire spreads by lognormal smoke and flame crossing times, gated by the barrier states, until the earliest of occupant control, fuel-limited burnout, or fire-service suppression. Detection depends on alarm state and on whether occupants are asleep; evacuation is simulated per occupant with heterogeneous awareness and pre-movement times, walking the shortest currently-tenable route to an exit and re-routing as nodes become untenable. Evacuation failure is a route cut before egress. The construction is a deliberately transparent stochastic caricature calibrated to plausible orderings and magnitudes, not a CFD or Fire Dynamics Simulator (FDS) replacement; every parameter is declared with a provenance tag in a single assumptions registry, and the physical orderings it encodes — better compartmentation and working detection lower every consequence probability, night and mobility impairment raise evacuation failure — are what make it *mechanism-*

informed and physically constrained rather than a black-box fit [Yarmohammadian et al., 2025, Yıldız, 2025].

Algorithm 1 states one Monte Carlo iteration exactly as implemented (`src/sim/fire.py` and `evacuation.py`); a cell probability is the sample mean of the recorded outcome over N such iterations (`run_experiments.py`, N = 10,000).

Algorithm 1 One iteration of the building fire–outcome simulator (one ignition).

Input: archetype graph $G = (V, E)$ (typed nodes; edges with barrier class b and length ℓ); scenario (alarm $\in$ {none, present-not-working, working}, time $\in$ {day, night}); RNG.
ignition $\leftarrow$ node sampled $\propto$ room-kind ignition weights (kitchen-dominant; exits excluded) ▷ `sample_ignition`
$(r_0, f_0) \leftarrow$ room and floor of the ignition node
for each edge $(u, v) \in E$ with barrier b **do**
  $p_{\text{open}} \leftarrow$ baseline open-probability of b; **if** night and b = internal door **then** $p_{\text{open}} \times= 0.35$
  open $\leftarrow [\mathcal{U}(0,1) < p_{\text{open}}]$; pick smoke, flame medians s_b, f_b for that open/closed state
  smoke time $t^s_{uv} \sim \text{Lognormal}(\ln s_b, \sigma{=}0.5)$; flame time $t^f_{uv} \sim \text{Lognormal}(\ln f_b, \sigma{=}0.5)$
end for
$\tau^s \leftarrow \textsc{Dijkstra}(\text{smoke times}, \text{ignition})$; $\tau^f \leftarrow \textsc{Dijkstra}(\text{flame times}, \text{ignition})$ ▷ additive shortest-path arrival
$t_{\text{det}} \leftarrow \textsc{Detection}(\tau^s, \text{alarm}, \text{night})$ ▷ working alarm: nearest origin-floor detector + lag; present-not-working: with prob. 0.15 a partial activation (detector + $\text{Lognormal}(\ln 120, 0.4)$ lag), else as no alarm; no alarm: occupant smoke cue at true occupant location + notice lag (long if asleep)
if $\mathcal{U}(0,1) < 0.74$ **then** ▷ minor fire: self-limits at the item first ignited
  $t_{\text{stop}} \sim \text{Lognormal}(\ln 120, 0.4)$; only room-$r_0$ nodes can become untenable (τ^s+ tenability lag); spread flags $\leftarrow$ False
else ▷ developing fire
  $t_{\text{stop}} \leftarrow \min\big($burnout; t_{det}+occupant control (w.p. 0.65 if an awake, able responder is present); t_{det}+call+FRS respo
  for each v: untenable$(v) \leftarrow \tau^s(v) + \text{Lognormal}(\text{lag})$ if $\tau^s(v) \leq t_{\text{stop}}$, else ∞
  burned $\leftarrow \{v : \tau^f(v) < t_{\text{stop}}\}$; spread-beyond-room/floor $\leftarrow$ any burned node outside r_0/f_0
end if
assign occupants to nodes (bedrooms at night; per-archetype load and mobility-impaired fraction)
for each occupant o **do**
  $a \leftarrow$ awareness time (τ^s at o's node + notice lag; roused by alarm if working); $t \leftarrow a +$ pre-movement delay (lognormal; + asleep, ×2.5 if impaired)
  if untenable$(o) \leq t$ **and** o asleep **then**
    mark FAILURE (overcome in place); **continue**
  end if
  repeat ▷ $\leq$ 40 re-route iterations
    $P \leftarrow$ min-travel-time path to any exit over nodes tenable now, skipping blocked nodes (walking speed reduced on stairs / in smoke)
    if $P = \varnothing$ **then** mark FAILURE (trapped); **break**
    walk P node by node; if the next node turns untenable before arrival, block it and re-plan; on reaching an exit record success and egress margin
  until exit reached or failed
end for
return spread-beyond-room, spread-beyond-floor, evacuation failure ($\geq$ 1 occupant fails), egress margins

Parameter provenance. Every constant lives in a single registry (`src/sim/params.py`) with a source tag, and the assumptions table in `results/sim/summary.md` is generated from it. The tags are: *data* — loosely calibrated to the `dwelling_fires` incident microdata; *PD7974* — order-of-magnitude values consistent with the PD 7974-6 / SFPE RSET decomposition; *lit* — qualitatively grounded in the review literature; *eng* — transparent engineering estimates chosen for plausible *ordering*, not fitted. Concretely: the ignition room-kind weights, the minor-fire fraction (0.74, calibrated so $\sim$85% of fires stay in the room of origin), and the suppression-chain call delay and FRS response time (median $\approx$8.5 min) are *data*-tagged. The tenability lag (median 90 s; 60 s at origin), the awake cue-notice and pre-movement times, and the able walking speed and stair factor (1.2 m/s, half speed on stairs) are *PD7974*. The asleep rousing penalty, the impaired walking speed (0.5 m/s), and the impaired pre-movement multiplier are *lit*. Everything else — crucially the smoke and flame edge-crossing medians per barrier class and their shared lognormal shape $\sigma = 0.5$, the barrier open-probabilities and the night door-shut factor, the detection and burnout medians, the smoke-slowing factor, and the occupant-control probability — is *eng*: these are engineering estimates set to reproduce the observed *orderings* (better compartmentation, working alarms and daytime all lower consequence probabilities), *not* CFAST outputs and *not* fitted. Because they are assumed rather than calibrated, we screen the conclusions against them by Morris elementary-effects analysis (Section 5.1) rather than assert robustness; the constants are therefore assumptions, sensitivity-checked. (The physics-calibrated t^2/CFAST engine of the companion paper is a separate model and does not feed these constants.)

What is and is not modelled. Tenability is a *single* criterion: a node becomes untenable a lognormal lag after smoke arrives, the lag standing in for the local ASET margin. Temperature, optical density/visibility and toxic (FED) dose are therefore *not* computed separately — the lag lumps them, and no ISO 13571 fractional-effective-dose integral is evaluated. There is likewise no heat-release-rate curve: flashover is not represented as a threshold event; "spread beyond the room/floor of origin" emerges purely from whether flame crossing times reach a compartment before the suppression horizon t_{stop}. Fuel packages are not itemised — the minor-fire branch represents fuel-limited item fires that self-extinguish, and the burnout time (median 1500 s) represents fuel-limited self-halt of a developing fire. Door leakage is represented implicitly: a closed barrier keeps a finite (long) crossing time rather than an infinite one, and each edge's open/closed state is drawn per iteration from the barrier's open-probability.

Running the simulator over a grid of archetype $\times$ alarm $\times$ time cells yields Monte Carlo consequence probabilities, which we compress into three interpretable logistic-regression surrogates with bootstrap prediction intervals. These surrogates are the closed-form $\mathbb{E}[C \mid I]$ engine consumed by the risk model of Section 3; their fit and behaviour are reported in Section 6.4.

5.1 Sensitivity of the conclusions to the engineering assumptions

Because the *eng* constants are assumed rather than fitted, we ask directly which of them the paper's conclusions depend on, by Morris elementary-effects screening [Morris, 1991, Campolongo et al., 2007]. Ten knobs — the global scale of the barrier crossing-time medians, their lognormal spread, the night door-shut factor, the minor-fire fraction, the burnout time, the occupant-control probability, the smoke-slowing factor, the tenability lag, the fire-and-rescue response time, and the pre-movement time — are each varied over a documented plausible range (roughly a factor of two either side of centre). We run 20 Morris trajectories on a six-level grid (220 design points, 4,000 Monte Carlo runs per archetype-by-alarm cell), and for each design point record two load-bearing outputs: the *interventional* alarm effect — the archetype-mean difference in evacuation-failure probability between an absent and a working alarm, the de-confounded quantity the simulator exists to supply — and the absolute failure level at a working alarm. Elementary effects are computed in normalised parameter space, so the importance measure μ^* (mean |effect|) and the interaction/nonlinearity measure σ (their spread) are comparable across knobs (Figure 1).

Steps are taken in the balanced deterministic-direction Morris variant, in which $\delta = p/(2(p-1))$ maps the lower half of the grid bijectively onto the upper half, so every level is sampled equiprobably and μ^* is unbiased; robustness of the ranking is assessed against a nominal design point with every constant at its central value.

Two conclusions are invariant to the assumptions. First, the interventional alarm effect stays *protective in all* 220 *perturbations* (range 0.03–0.19, median 0.09; it never changes sign), so the mechanism the simulator contributes to the paper — that detection lowers consequence, the effect the observational data cannot identify — does not rest on any particular constant. Second, the seven-archetype ordering is essentially fixed: the Kendall rank correlation against the nominal ordering is at least 0.90 at every design point, with the care-home archetype the worst in every one. What the assumptions *do* move is the magnitude, and for both outputs that movement concentrates in a few near-linear directions ($\sigma < \mu^*$) with the remaining knobs sloppy. For the interventional alarm effect the stiff directions are the minor-fire fraction ($\mu^* = 0.086$), the barrier crossing-time scale (0.035) and the burnout time (0.028), together two-thirds of the total; for the absolute failure level they are the minor-fire fraction (0.032) and the crossing-time scale (0.025) with the occupant-control probability third (together about three-fifths), burnout falling among the sloppy knobs there. This is the same signature the companion paper finds in the physics calibration, where aggregate incident data identify essentially one stiff direction and leave the rest prior-dominated: in both models a few directions are stiff and most are sloppy, which is why we report a de-confounded *contrast* and an archetype *ranking* rather than absolute casualty counts. The screening is reproducible via `python -m src.sim.morris_sweep`.

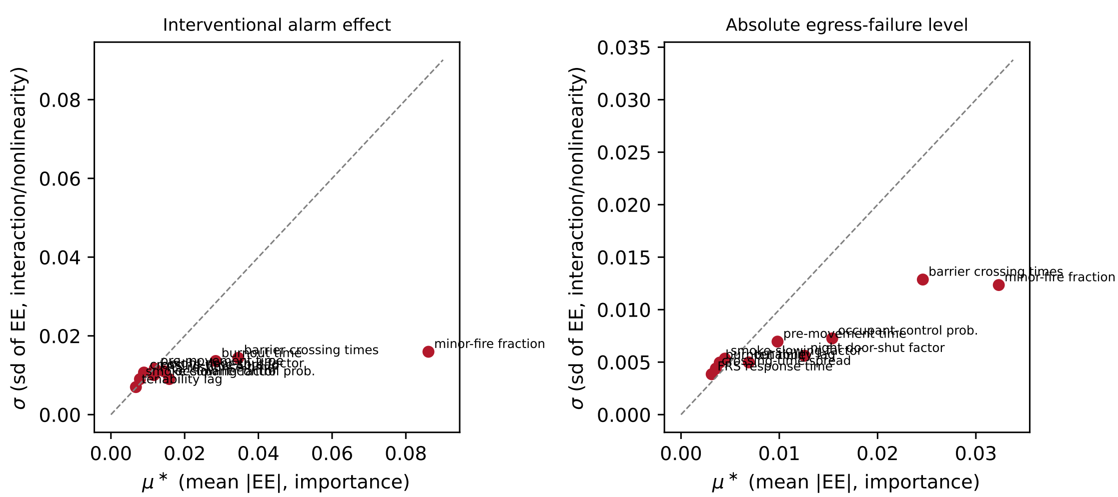

Figure 1: Morris elementary-effects screening of the ten *eng* simulator assumptions, for the interventional alarm effect (left) and the absolute evacuation-failure level (right). Each point is one assumption; μ^* (horizontal) is its importance and σ (vertical) its interaction/nonlinearity. The minor-fire fraction and crossing-time scale are the stiff directions for both outputs; burnout time is additionally stiff for the alarm effect (left) but sloppy for the absolute level (right). The interventional alarm effect stays protective and the archetype ranking fixed (Kendall $\tau \geq 0.90$) across all 220 design points.

6 Results

We instantiate the framework of Section 3 in three fitted components — a spatial occurrence model ($P(I)$), record-level consequence models ($P(S \mid I)$, $P(\text{casualty} \mid I)$), and the archetype simulation surrogate ($P(E_f \mid S, I)$) — and then assemble them into the risk decomposition of Equation (1). Every number below is a held-out or out-of-time evaluation; in-sample fit is not reported.

6.1 Fire occurrence

We model dwelling-fire counts per LSOA-year as a hierarchical negative-binomial (NB) regression over all 33,755 English LSOAs, with a household-count exposure offset, ten standardised covariates, a linear time trend, and a random intercept for each of the 44 FRS areas. Full No-U-Turn Sampler (NUTS) sampling on the 202,530-row training frame is infeasible on CPU: the step size collapses against a stiff posterior geometry under non-centred, centred and QR-orthogonalised parameterisations alike (to 5×10^{-7} in the best case). The exact full-data posterior is nevertheless recovered without sampling, by Pareto-smoothed importance sampling over a Laplace proposal (Section 3): the smoothing diagnostic $\hat{k} = 0.33$ certifies the correction, with an importance-sampling effective sample size of ∼16,900 from 50,000 draws, and the result agrees with an exact NUTS fit on a stratified subsample of ∼10,000 LSOAs, full-data ADVI and full-data MLE. Calibration is evaluated on the full held-out population (all 33,755 LSOAs). Training uses FY 2017/18–2022/23; FY 2023/24 and 2024/25 are held out.

That the Laplace-plus-PSIS construction recovers the *posterior*, and not merely a well-behaved proposal, rests on the posterior being close to the Gaussian the Laplace step assumes — which we verify directly rather than take on faith from $\hat{k}$ alone. Across all 58 unconstrained coordinates the importance-weighted posterior is empirically near-Gaussian: the maximum absolute skewness is 0.13 (median 0.04) and the maximum excess kurtosis 0.33 (median 0.07), the mild departures Bernstein–von Mises predicts at roughly 3,500 observations per parameter (202,530 rows over 58 coordinates). A multimodality probe finds no competing mode — six L-BFGS optimisations from widely dispersed starts (perturbation SD 2 in the unconstrained space) all converge to a single optimum, agreeing to 3×10^{-5} in objective value and 8×10^{-4} in location. The honest caveat sits at the tail of the hierarchy: rows per fire-and-rescue service range from 6 (Isles of Scilly) to a median of 3,444, so the smallest service effects remain partially prior-informed — yet their marginals still fall inside the skewness bound above, and this partial pooling is the intended shrinkage of the hierarchical model, not a failure of the asymptotics. Finally, $\hat{k}$ is itself a sensitive detector of proposal–posterior mismatch — heavy tails or a missed mode inflate it — so $\hat{k} = 0.33$ corroborates the near-Gaussian geometry established above rather than serving as its sole warrant.

Because dwelling-fire counts per LSOA-year are small (mean ≈ 0.8, mostly 0–2), raw central-interval coverage structurally over-covers; the correct discrete-count calibration notion is the randomised probability integral transform (PIT), which is $\text{Uniform}(0, 1)$ under calibration. On that metric the negative-binomial model is well calibrated and beats an equal-mean Poisson on both held-out years: from the exact full-data posterior, randomised coverage of 0.504 / 0.901 against nominal 0.50 / 0.90 in 2023/24 (and 0.495 / 0.899 in 2024/25), with a PIT Kolmogorov–Smirnov (PIT–KS) distance of 0.0045 versus the Poisson's 0.0145 (subsample track). It also improves markedly on an intercept-only null in held-out log predictive density (-1.101 vs -1.173 per LSOA-year in 2023/24), confirming that the covariate and service structure carry genuine predictive signal and that overdispersion beyond Poisson is real, if moderate at this aggregation. Figure 2 shows the reliability curve and PIT histogram.

Covariate effects (Table 2, exact full-data posterior) are reported as rate ratios per one-standard-deviation (SD) increase. Overall deprivation dominates: a one-SD rise in the IMD score raises the dwelling-fire rate by 32% (rate ratio 1.322, 95% credible interval (CrI) [1.310, 1.334]). Building form (% flats, 1.117) and the health-deprivation domain (1.086) are the next strongest signals; % privately rented carries the weakest signal of any covariate (1.014 — its interval now excludes 1 at full-data precision, but the effect is negligible against deprivation's 32%) once deprivation and form are controlled. The domain coefficients are *conditional* on overall IMD (with which they are collinear, e.g. $r \approx 0.81$ for the health domain) and should not be read as marginal effects. This spatial gradient reproduces the established relationship between deprivation and fire incidence (Figure 3): predicted and observed dwelling-fire rates fall monotonically from roughly 2.1 to 0.6 fires per 1,000 households across deprivation deciles. Calibration is slightly worse in 2024/25 (PIT–KS 0.029): the linear trend estimated through 2022/23 does not fully capture that year's steeper decline, so the model marginally over-predicts it — an honest forward-forecast limitation, with coverage still on target.

Table 2: Occurrence-model covariate effects: rate ratio per +1 SD, with 95% credible interval; exact full-data posterior (Laplace + PSIS, all 202,530 training rows). Deprivation is the dominant driver.

Covariate	Rate ratio	95% CrI
IMD score (overall deprivation)	1.322	[1.310, 1.334]
% flats (Census)	1.117	[1.109, 1.126]
Health-deprivation domain	1.086	[1.075, 1.097]
% overcrowded	1.047	[1.039, 1.055]
Living-environment domain	1.044	[1.034, 1.054]
% lone-person aged 66+	1.043	[1.037, 1.050]
% dwellings pre-1919 (EPC)	1.027	[1.017, 1.037]
% converted flats	1.025	[1.019, 1.031]
% electric main heat	1.017	[1.009, 1.024]
% privately rented	1.014	[1.005, 1.022]

Robustness (full-data, ablation, spatial, subgroups). Three checks confirm the model. The exact full-data posterior above is corroborated by an independent full-data ADVI refit, which reproduces every rate ratio to within 0.022 and *strengthens* the overdispersion result — the equal-mean Poisson undercovers badly (50% interval covers 0.401) while the negative binomial stays on target (0.503 / 0.901). An ablation ladder (Table 3) shows deprivation is by far the largest single gain, the EPC covariate adds essentially nothing at LSOA level, and a Besag–York–Mollié (BYM2) spatial random effect removes the residual spatial autocorrelation the service intercepts leave (Moran's I falls $0.059 \rightarrow -0.086$) for only a small held-out gain — real but modest, and this M5 specification is preferred over the rejected quadratic-year M6. And calibration holds *within* subgroups: held-out PIT–KS stays near 0.01 across IMD quintiles, urban/rural strata and %-flats tertiles, with median 0.039 across the 43 service areas.

6.2 Consequence severity

Two record-level outcomes are modelled from the 451,172 dwelling-fire records by hierarchical Bayesian logistic regression, each with an FRS-level random intercept. *Serious spread* is fire leaving the room of origin (a direct proxy for compartmentation failure); *any-casualty* is the data-protected binary casualty indicator (fatalities cannot be separated from non-fatal casualties at record level). The models are fit by *exact* NUTS on all 451,172 records — no subsampling — trained on FY

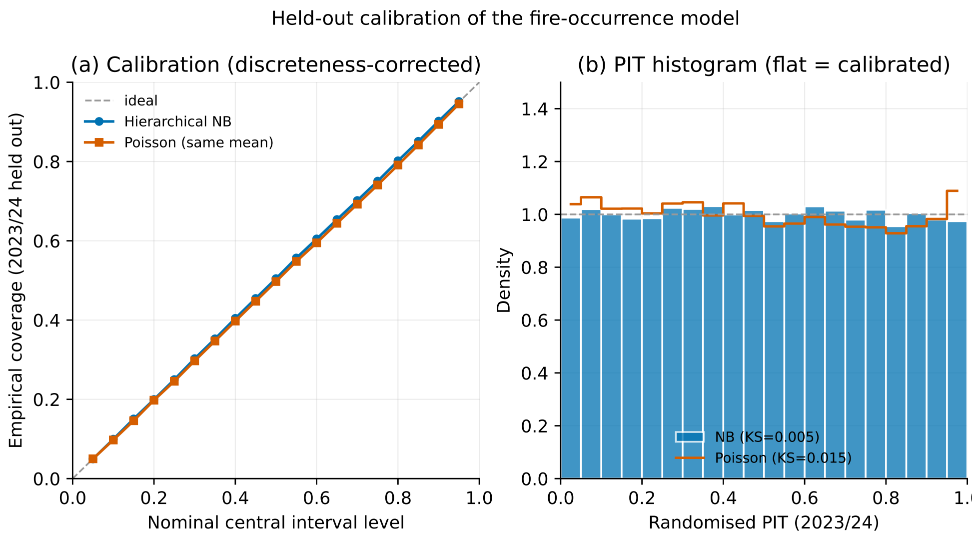

Figure 2: Held-out calibration of the occurrence model (FY 2023/24): coverage of central posterior-predictive intervals (left) and randomised-PIT histogram (right). The PIT is close to Uniform, indicating good discrete-count calibration.

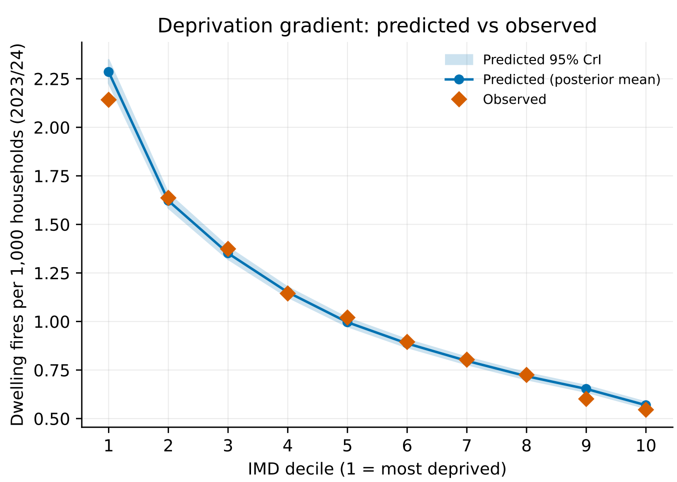

Figure 3: Dwelling-fire rate per 1,000 households by IMD decile, predicted vs observed. The monotone deprivation gradient is reproduced. Covariate rate ratios are in Table 2.

Table 3: Occurrence ablation ladder (full-data variational fits, so M5 point estimates differ slightly from the exact posterior of Table 2), held-out 2023/24. Deprivation dominates; the EPC covariate adds nothing at LSOA level; the intrinsic-conditional-autoregressive (ICAR)-based BYM2 spatial effect is real but modest; the preferred model is M5. Log-score is NB held-out log predictive density (higher is better).

Rung	Log-score	PIT–KS	cov50	cov90
M0 intercept + trend	−1.1725	0.0056	0.506	0.902
M1 + deprivation	−1.1155	0.0047	0.504	0.900
M2 + building form (Census)	−1.1024	0.0044	0.504	0.902
M3 + EPC covariate	−1.1024	0.0045	0.503	0.902
M4 + service random effect	−1.1000	0.0039	0.502	0.901
M5 + BYM2 spatial (preferred)	−1.0976	0.0053	0.495	0.898
M6 + quadratic year (rejected)	−1.0963	0.0146	0.505	0.904

2010/11–2022/23 ($n = 400{,}250$) and evaluated out-of-time on 2023/24–2024/25 ($n = 50{,}922$), with every reported fixed effect converged cleanly ($\hat{R} \leq 1.02$, zero divergences). A fiscal-year-stratified subsample fit ($n = 29{,}999$) and an independent maximum-likelihood (MLE) fit on the full records serve as cross-checks under the two-track protocol of Section 3, reproducing the same effect signs and comparable magnitudes. The current specification adds an FRS-year *response-time context* covariate — the mean total call-to-arrival delay per FRS and fiscal year from FIRE1001, joined exactly to every incident (0% unmatched) and standardised (1 SD $\approx$ 1.23 minutes about a 7.71-minute mean). Both models are well calibrated out-of-time: expected calibration error (ECE) is 0.0096 for spread and 0.0108 for casualty, and held-out area under the receiver-operating-characteristic curve (AUC) is 0.752 and 0.685 respectively (Figure 5); per-service interval coverage is discussed under robustness below.

Response time itself carries only a weak, non-robust signal and must be read as an *ecological* exposure, not an incident-level measurement: it varies only at FRS$\times$year resolution, so with the FRS random intercept absorbing between-area differences it is identified largely from within-FRS year-to-year variation and is partly entangled with the secular fiscal-year trend. On the subsample the casualty odds ratio (OR) per +1 SD is 0.851 (95% CrI [0.777, 0.931]) and the spread OR 0.922 ([0.851, 0.993]); but on the full 451,172 records the spread effect *flips sign* (OR 1.052, [1.018, 1.084]) while casualty stays below one (0.944, [0.915, 0.973]). We therefore report no directional response-time effect on spread, and read the (still ecological, non-causal) casualty association only weakly; the covariate is retained as a control, not a finding.

The headline effect is the alarm ladder (Table 4). For serious spread the signal is clean and mechanistically sensible: relative to a working alarm, a fire with *no* alarm has 1.65 times the odds of spreading beyond the room of origin (OR 1.653, 95% CrI [1.610, 1.693]), and an alarm present that did not raise the alarm carries the highest odds of all (OR 2.401). Detection cuts spread. The casualty model, however, shows the no-alarm odds ratio falling *below* one (OR 0.786, [0.766, 0.804]). This sign flip is not an alarm being harmful; it is the expected observational confound. A dwelling fire only produces a casualty when someone is present and the fire is serious, and those same conditions are what make an installed alarm sound — so alarm operation is correlated with exactly the occupancy and severity that raise casualty risk. Figure 4 draws this backdoor path explicitly. The record-level data cannot by itself identify the causal casualty-prevention effect; it identifies adjusted associations. We report both outcomes precisely because the contrast — a clean detection$\to$spread signal alongside a confounded casualty association — is the motivating case for the latent-state layer of Section 3, whose job is to de-confound occupancy from detection. Reassuringly, the alarm ladder is essentially unchanged by adding the response-time covariate (assessed on the subsample track: the no-alarm spread OR moves 1.699→1.701 and the no-alarm casualty OR is unchanged) while calibration slightly improves, so the alarm signal is robust to the new covariate. Other effects are large and interpretable: chip/fat-pan fires multiply casualty odds by 3.25, lone pensioners by 2.21, and night ignition by 1.83; purpose-built high-rise flats have the lowest spread odds of any dwelling type (OR 0.378 vs houses), consistent with their compartmentation.

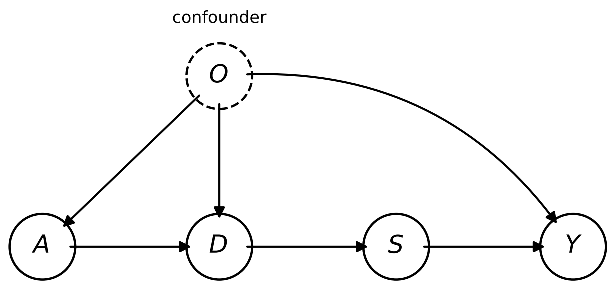

Figure 4: Causal structure behind the alarm–casualty association. Occupancy O (dashed, the confounder) opens the backdoor path $A \leftarrow O \rightarrow Y$: occupied, well-maintained dwellings both carry a working alarm ($O \rightarrow A$) and contain someone who can be harmed ($O \rightarrow Y$), and present occupants themselves aid detection ($O \rightarrow D$), while the alarm reaches casualties only along the causal chain $A \rightarrow D \rightarrow S \rightarrow Y$ (alarm A, detection D, severity/spread S, casualty Y). An adjusted no-alarm casualty odds ratio below one is the expected signature of this structure — alarm operation tracks the very occupancy that raises casualty risk — rather than a protective effect of *removing* alarms. The record-level data identify adjusted associations along this graph, whereas the intervention effect requires the do-operator treatment referenced in Section 3.

Table 4: Alarm ladder: odds ratios (95% CrI) for each alarm state versus a working alarm (present and raised), adjusted for dwelling type, cause, ignition, occupancy, time, year, response time and FRS; exact NUTS posterior on all 451,172 records. Detection lowers spread; the casualty column is confounded by occupancy (see text).

	Serious spread		Any casualty	
Alarm state vs. working	OR	95% CrI	OR	95% CrI
Absent	1.653	[1.610, 1.693]	0.786	[0.766, 0.804]
Present, did not operate	1.116	[1.083, 1.148]	0.789	[0.769, 0.810]
Present, did not raise alarm	2.401	[2.329, 2.474]	1.196	[1.162, 1.230]

Robustness (proper scores, full data, subgroups). Because the area under the ROC curve alone is a weak test for rare outcomes, we also report strictly proper scores (Table 5): the hierarchical model beats an intercept-only baseline decisively on the Brier score, mean log score and precision–recall AUC (PR-AUC), matches a plain logistic while adding calibrated uncertainty, and sits well above the PR-AUC base-rate floors (≈ 0.15 / 0.13). The exact full-data posterior agrees with the

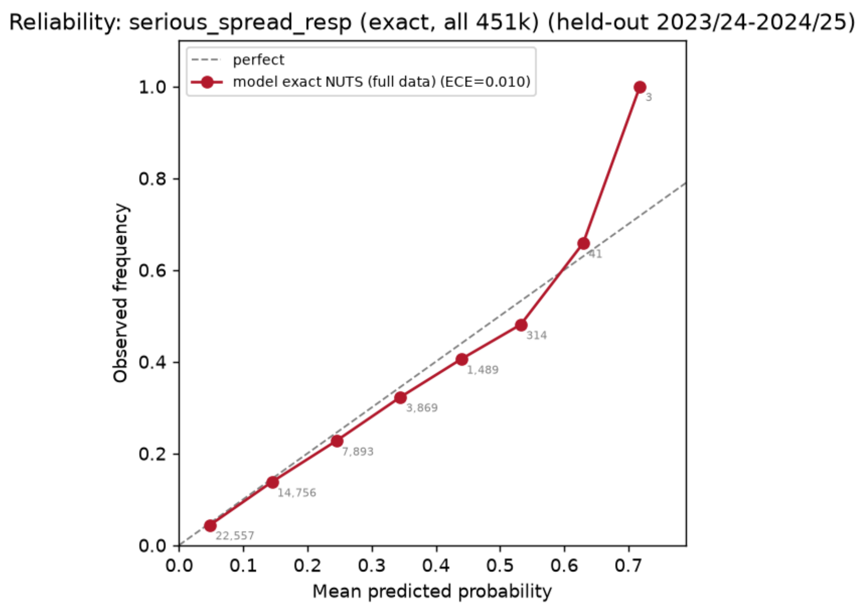

Figure 5: Out-of-time reliability curve for the serious-spread model (exact full-data NUTS, response-time specification, FY 2023/24–2024/25), ECE 0.0096. Predicted probabilities track observed frequencies across bins.

subsample track, full-data ADVI and full-data MLE on every headline effect (no-alarm casualty OR 0.758 → 0.786, spread 1.70 → 1.65), and its ∼4× tighter intervals resolve two subsample artefacts: the deliberate-cause casualty OR collapses from a wide, nominally significant 3.04 [1.05, 10.6] to a clean null (1.12, [0.92, 1.38]), and the response-time spread coefficient settles at the sign discussed above. One honest caveat surfaces at scale: the full-data fits undercover per-FRS held-out counts (0.81 for spread and 0.65 for casualty under exact NUTS, versus 1.00/0.93 on the subsample), and since the *exact* sampler undercovers too this is a real FRS-overdispersion gap (out-of-time drift and within-service heterogeneity a single intercept misses), not a variational artefact — fixed effects are best estimated from the full data, while per-service predictive intervals from the tighter full-data posterior should be read as anti-conservative and a random-slope extension is future work. Calibration broadly holds *within* alarm, dwelling-type and time-of-day subgroups (most subgroup expected calibration errors of the same 0.01–0.02 order as the marginal figure, with larger wobbles at small-n cells such as bungalows, ECE 0.09 at n=189).

Table 5: Held-out proper scores for the consequence models versus an intercept-only baseline (subsample held-out, n = 3,386). Lower Brier and log score are better; higher PR-AUC and ROC-AUC are better. The hierarchical model ranks rare events well above the base-rate floor.

Outcome	Model	Brier	Log score	PR-AUC	ROC-AUC
Casualty	Hierarchical	0.1205	0.3984	0.245	0.659
Casualty	Intercept-only	0.1250	0.4166	0.146	0.500
Serious spread	Hierarchical	0.1041	0.3446	0.318	0.757
Serious spread	Intercept-only	0.1160	0.3939	0.134	0.500

6.3 Interaction structure: who gains most from an alarm

Does the alarm benefit differ by occupancy, and the night penalty by dwelling type? We add two interactions to the consequence model — alarm×occupancy and night×dwelling-type — and, because interaction cells are noisy, hold them to a strict discipline: a joint likelihood-ratio (LR) test of each interaction block on the full data, Benjamini–Hochberg false-discovery-rate (BH-FDR) control across cells, and a dual gate that calls a cell robust only if it clears *both* the Bayesian 95% credible interval and the full-data maximum-likelihood interval with the same sign. Of 68 tested cells, the robust ones are 8 (spread) and 5 (casualty) in alarm×occupancy and 2 (casualty) in night×dwelling.

The spread channel — the cleaner, less-confounded detection signal — shows a genuine *occupancy gradient* in the value of a working alarm (Figure 6). The no-alarm-versus-working spread odds ratio runs from about 3.30 [3.04, 3.57] for a lone person over pensionable age — roughly double the pooled 1.65 — down to indistinguishable-from-none for families with children (who have waking occupants to detect a fire). Lone pensioners, lone under-pensioners and older couples all sit robustly above the pooled penalty (the cell probabilities are reported as adjusted average marginal effects, AME). By contrast the night effect on spread is a *whole-stock* phenomenon: none of the seven dwelling deviations from the house reference clears both gates, so no building form — HMOs and converted flats included — is disproportionately worse at night once the shared night penalty is removed. The interaction terms do not degrade calibration (held-out ECE 0.012–0.014). The targeting consequence is drawn out in Section 7: the alarm benefit should be occupancy-modulated toward lone pensioners, while night warrants no dwelling-specific multiplier.

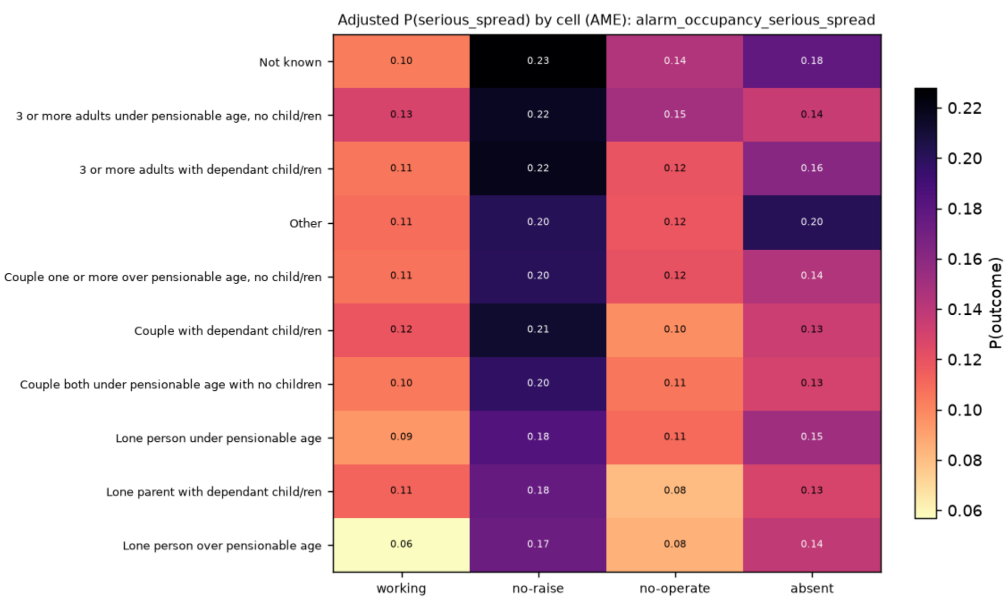

Figure 6: Adjusted probability of serious spread by occupancy type (rows) and alarm state (columns), from the alarm×occupancy interaction. The no-alarm spread penalty is largest for lone pensioners, who gain the most containment from a working alarm.

6.4 Simulation and surrogate

The evacuation-failure factor $P(E_f \mid S, I)$ comes from the archetype simulator of Section 5, run for 420,000 Monte Carlo ignitions (10,000 per archetype × alarm × time-of-day cell, fixed seed). Every parameter is declared in a single assumptions registry with a provenance tag (calibrated to data, consistent with PD 7974 and Society of Fire Protection Engineers (SFPE) practice, grounded in the literature, or a transparent engineering estimate), so the model is auditable rather than a black box. Table 6 gives the headline night-time per-occupant evacuation-failure probabilities.

Table 6: Per-occupant P(evacuation failure | ignition) at night, by archetype and alarm state (Monte Carlo, 95% confidence intervals (CIs) in the full results). A working alarm roughly halves failure in most archetypes; the care home is the exception (see text).

Archetype	no alarm	alarm not working	working alarm
detached house	0.385	0.370	0.249
terraced house	0.404	0.396	0.242
low-rise flat	0.141	0.132	0.049
high-rise flat	0.071	0.068	0.012
HMO	0.364	0.355	0.262
care home	0.163	0.165	0.168
mixed use	0.181	0.176	0.070

The care home is the informative exception: its evacuation-failure rate barely moves with alarm state (0.163 / 0.165 / 0.168 at night), because its bottleneck is occupant *mobility*, not *detection* — a 75% mobility-impaired population with long pre-movement times cannot exploit earlier warning under a full-evacuation assumption. Detection helps where egress is fast (flats, houses); it does not where egress is slow. This is exactly the kind of archetype-specific mechanism the surrogate is meant to expose.

Against the observed dwelling-fire statistics the simulator is *loosely* calibrated — we target orderings and magnitudes, not exact matches — and the orderings hold. Spread beyond the room of origin ranks working alarm < not working < none (simulation 0.100 / 0.185 / 0.184 vs data 0.079 / 0.146 / 0.185) and night > day (0.176 vs 0.137, data 0.207 vs 0.117); the compartmentation ordering is strong, with purpose-built flats almost never spreading beyond the floor of origin. The one acknowledged discrepancy is that multi-storey *houses* over-predict spread beyond the floor (simulation $\approx$0.13 vs data 0.05), because an open internal stair lets fire climb too readily; flats match closely. The simulator also independently reproduces the same casualty confound seen in the record data — observed casualty rate is *highest* with a working alarm — while its causal $P(E_f)$ correctly *decreases* with a working alarm, which is the interventional quantity the risk model needs.

The new building-type benchmarks (FIRE0301/0304) give the care-home and mixed-use archetypes their first real calibration targets, and they validate the institutional archetypes with no change to the simulator's parameters. The care-home archetype's spread-beyond-room rate (0.079) matches the observed 'communal living' class almost exactly (0.082) and sits far below the commercial classes (retail 0.203, food-and-drink 0.225), confirming its institutional compartmentation is at the right level. On consequence, communal-living buildings have the highest casualty-per-fire of the residential-type classes (0.110), and the care-home archetype's per-occupant evacuation-failure rate (0.156) is the right order of magnitude, overstating the observed per-occupant casualty rate by roughly 1.4×; the building-level $P(\geq 1$ occupant fails) of 0.528 overstates by construction far more (∼4.8×) — both are the no-staff-assist, stay-put- and sprinkler-free full-evacuation upper bound flagged as a limitation, now quantified against data. Separately, the smoke-alarm operation rate $P(\text{operated} \mid \text{present}) = 0.743$ (FIRE0703) and working-alarm ownership 0.924 (FIRE0701) confirm the *structure* of the three-way alarm split — both a large “working” and a large “present-but-failed” population — but they are timing-free aggregate counts, so no detection parameter was changed.

Finally, three logistic-regression surrogates map archetype features (storeys, occupants, compartmentation, impaired fraction, alarm state, night) to the three consequence probabilities, recovering the Monte Carlo cell probabilities with $R^2 = 0.861$ (spread beyond room), 0.969 (spread beyond floor) and 0.936 (evacuation failure), all with interpretable coefficient signs and bootstrap prediction intervals. This surrogate is the closed-form $\mathbb{E}[C \mid I]$ engine that plugs into Equation (3) (Figure 7).

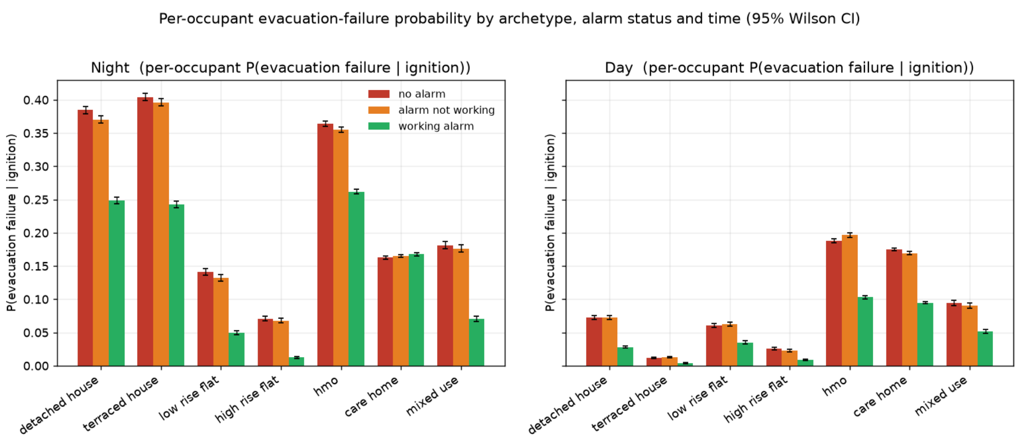

Figure 7: Per-occupant probability of evacuation failure by archetype, alarm state and time of day, with 95% Monte Carlo error bars. A working alarm lowers failure in every archetype except the care home, whose bottleneck is mobility, not detection.

6.5 Synthesis: a worked risk decomposition

The three components combine into the posterior risk of Equation (1), carrying uncertainty through each factor. Consider an illustrative high-rise flat in a most-deprived-decile LSOA at night with no working alarm. The occurrence model places the annual ignition probability near the top of the deprivation gradient, $P(I) \approx 2.1 \times 10^{-3}$ per household-year (Figure 3); the surrogate gives, for the high-rise-flat archetype with no alarm at night, $P(S \mid I) \approx 0.24$ and $P(E_f \mid S, I) \approx 0.071$. Composing these,

$$R_b \;=\; \underbrace{P(I)}_{\approx 2.1\times 10^{-3}} \cdot \underbrace{P(S \mid I)}_{\approx 0.24} \cdot \underbrace{P(E_f \mid S, I)}_{\approx 0.071} \cdot \mathbb{E}[L \mid \cdot] \;\approx\; 3.6 \times 10^{-5} \cdot \mathbb{E}[L \mid \cdot] \text{ per hh-year}, \tag{21}$$

before the loss term. Fitting a working alarm collapses the middle factor toward $P(S \mid I) \approx 0.09$ and the egress factor to $P(E_f \mid S, I) \approx 0.012$, an order-of-magnitude reduction in the evacuation-failure contribution — the quantitative expression of the intervention. To collapse these alarm-conditional outputs into a single population-average consequence for a stock of mixed alarm reliability, the surrogate’s working and not-working predictions are combined using the empirical smoke-alarm operation rate $P(\text{operated} \mid \text{present}) = 0.743$ (FIRE0703; Section 6.4) as the population weight — supplied to the risk model rather than baked into the simulator. Crucially each factor carries a credible interval (the occurrence posterior, the consequence-model CrIs of Table 4, and the surrogate's bootstrap prediction intervals), so the product in Equation (21) is a *distribution* over R_b, not a point score, and the width of that distribution is itself the missing-evidence signal that drives inspection prioritisation (Section 3.4). These figures are illustrative of the composition rather than a calibrated building-level prediction; building-level validation remains at the aggregate level (Section 7).

6.6 Decision layer: value of targeted prevention

The framework exists to support decisions, so we close the analysis with a worked value-of-inspection demonstration (contribution 4): given a *fixed* prevention budget, how much better can a fire and rescue service target it using the calibrated posteriors than with untargeted or deprivation-only rules? This is emphatically *not* an argument against statutory fire risk assessment — PAS 79 [British Standards Institution, 2020a], the Building Safety Act and the post-Grenfell regime are valued safety improvements. The decision layer is a quantitative prioritisation layer that *backs* that regime with data: which homes, in which order, for a limited budget. The policy evaluated is a real intervention — FRS home fire-safety ("Safe and Well") visits that install or check a working smoke alarm — at a fixed budget of 100,000 visits across England. Four ranking rules are evaluated against the same world (Table 7): random, IMD-decile (current common practice), posterior fire rate, and a full expected-benefit rule that is uncertainty-aware and consequence-valued (rate $\times$ P(no working alarm) $\times$ P(casualty | fire) $\times$ relative risk reduction (RRR)).

Economic values are the Home Office *Economic and Social Cost of Fire* [Home Office, 2023]: averting one casualty-fire is worth £66,113 (national-average severity, dominated by the rare fatality) and a working alarm averts an assumed 20% of a fire's £14,009 property damage; benefits accrue over a 10-year alarm life discounted at 3.5% (present value (PV) multiplier 8.61) against a £55 visit cost. The alarm's causal effect is taken from the *interventional* surrogate rather than the confounded observational casualty odds ratio (which is below one), and is cross-checked against the observational spread signal; every causal assumption is tabled with a sensitivity sweep.

Targeting multiplies the return roughly ninefold: expected-benefit targeting returns £0.71 per £1 spent versus £0.078 for random allocation, and averts 3.1 casualty-fires per year against 0.34 (Table 7). The calibrated occurrence model also beats the deprivation proxy by about 2.5× (3.0 vs 1.2 casualty-fires averted for posterior-risk vs IMD-decile targeting): deprivation is a genuinely strong single proxy — it co-varies with fire rate, the alarm gap and vulnerability at once, which is why services use it and we treat it respectfully — but within the deprived deciles the calibrated model pinpoints the specific high-rate LSOAs that random-within-decile misses. The consequence and uncertainty-aware signals add a further modest gain (0.705 vs 0.673); they are collinear with the fire rate here, so once the rate is modelled well they add little, and they would matter more where consequence decouples from occurrence. The absolute return sits below £1/£1 by construction: we value *only* the alarm's consequence channel and exclude the visit's fire-prevention advice and health co-benefits, so every figure is a deliberate lower bound. The contribution is the targeting multiplier and the uncertainty-aware decision rule, not a claim that alarm installation pays for itself on casualties alone; a sensitivity sweep shows the break-even frontier crossing the plausible parameter region — reached around a £30 visit cost at the base effect size, and requiring correspondingly larger interventional effects at higher costs (Figure 8).

Formal allocation, value of inspection, and budget size. The allocation is the constrained maximisation of expected averted loss of Equation (13); because every dwelling in an LSOA shares the same marginal value, it is a continuous knapsack whose optimum is the greedy rule (rank by marginal value, fill from the top), and policy (d) is its Bayes action — ranking by posterior-mean

Table 7: Value of targeting a fixed 100,000-visit smoke-alarm budget under four ranking rules, evaluated against the same England-wide world. Present value over a 10-year alarm life at 3.5%, casualty plus property, with 90% credible intervals propagated from the occurrence posterior and surrogate bootstrap.

Policy	LSOAs	Casualty-fires averted/yr	£saved per £spent
(a) Random	33,755	0.34 [0.33, 0.35]	0.078 [0.076, 0.080]
(b) IMD decile	145	1.19 [1.15, 1.24]	0.274 [0.267, 0.282]
(c) Posterior risk	127	3.00 [2.87, 3.14]	0.673 [0.649, 0.697]
(d) Expected benefit	127	3.13 [2.99, 3.29]	0.705 [0.678, 0.731]

averted loss. Reframing the visit as an inspection that resolves alarm ownership gives the EVI of Equation (12), which sharpens the picture: the prior-optimal action is to visit in only 0.04% of dwellings (expected benefit exceeds the £55 cost almost nowhere), yet a visit *would* pay in the 29.0% of dwellings that turn out to lack a working alarm — so the programme's whole value lies in *finding* those homes, which is what inspection does, with total EVI over England $\approx$ £22.7m. EVI is also a genuinely different ranking from expected risk (Spearman $\rho = 0.79$, versus 0.985 for expected-benefit-versus-risk): it zeroes out the $\sim$71% of areas where a visit can never pay, so uncertainty-reducing targeting is distinct from risk targeting even though the two coincide on the top LSOAs a 100,000-visit budget reaches.

The headline economic result is a break-even one, and it is about concentration, not scale. Sweeping the budget (Table 8), a small, tightly targeted 10,000-visit programme clears break-even on the smoke-alarm channel alone (£1.12 per £1) even on our conservative casualty-plus-property valuation; the advantage over IMD targeting then falls from 4.4× at 10,000 visits to 1.4× at one million as a larger programme is pushed down into lower-risk stock. A decision-curve analysis and the visit-cost×effect-size sensitivity map (Figure 8) place the break-even frontier inside the plausible region: cheaper visits or the upper half of the effect-size range lift the channel above £1/£1.

Table 8: Return per pound (present value, casualty plus property) by programme size and policy. A small, tightly targeted programme clears break-even (£1.12); the targeting advantage over IMD decays with scale (classic diminishing returns).

Visits	Random	IMD decile	Posterior risk	Expected benefit
10,000	0.078	0.252	1.112	**1.119**
50,000	0.078	0.255	0.800	0.821
100,000	0.078	0.256	0.673	0.704
250,000	0.078	0.261	0.545	0.563
500,000	0.078	0.260	0.449	0.464
1,000,000	0.078	0.267	0.362	0.374

What finer geography buys: sub-LSOA concentration in London. The decision results above inherit the framework's granularity ceiling: the national incident data resolve only to LSOA. London is the one fire service that publishes incident records at finer resolution — dwellings on a 50 m grid — and a pilot on those records (90,318 dwelling fires, 2009–2024, with EPC-derived dwellings per cell as exposure) shows how much that ceiling costs. Within an average London LSOA, the top 10% of inhabited 50 m cells hold **83%** of the LSOA's dwelling fires, against 59% expected from dwelling density and small-count sampling noise alone (a dwelling-density multinomial null;

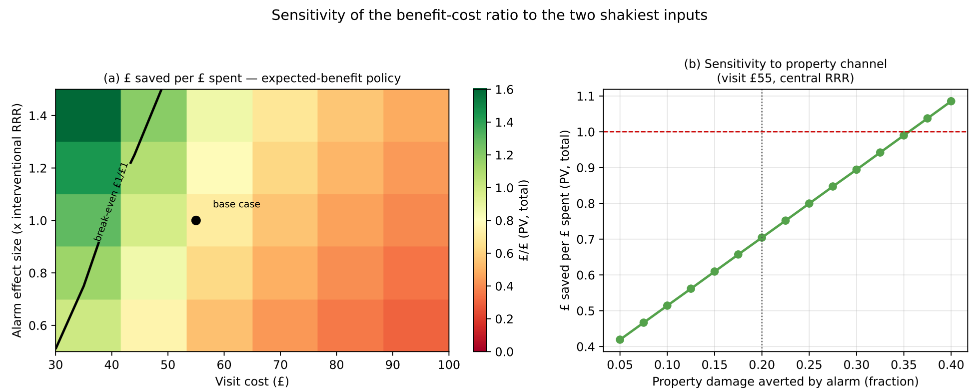

Figure 8: Return per pound over visit cost (£30–£100) and alarm effect size (0.5–1.5× the interventional RRR), with the break-even contour and base-case marker. The frontier crosses the plausible region.

the honest signal is the 24-point excess). Priced with the same economics as the policy comparison above, and evaluated *out-of-sample* (cells ranked on 2009–2020 by empirical-Bayes shrunk rates, every selected dwelling scored at its realised 2021–2024 rate), cell-level targeting reaches dwellings with **5.7**× the fire rate of LSOA-level targeting at a 10,000-visit London budget — lifting the smoke-alarm channel from £0.33 to £1.88 saved per £1 spent, i.e. above break-even where LSOA targeting is not close (4.2–5.0× at larger budgets; a perfect-foresight oracle reaches 11×, the gap measuring how much apparent concentration is noise). Both caveats push the estimate to be conservative: the 50 m rounding coarsens true building-level concentration, and the ∼14% of fires falling outside EPC-covered cells are dropped from observed and null alike (Figure 9). The implication is not that the national model is wrong — its calibration holds — but that its geography leaves roughly a sixfold value multiplier on the table, which is precisely the case for record-level access to fine incident geography (Section 7).

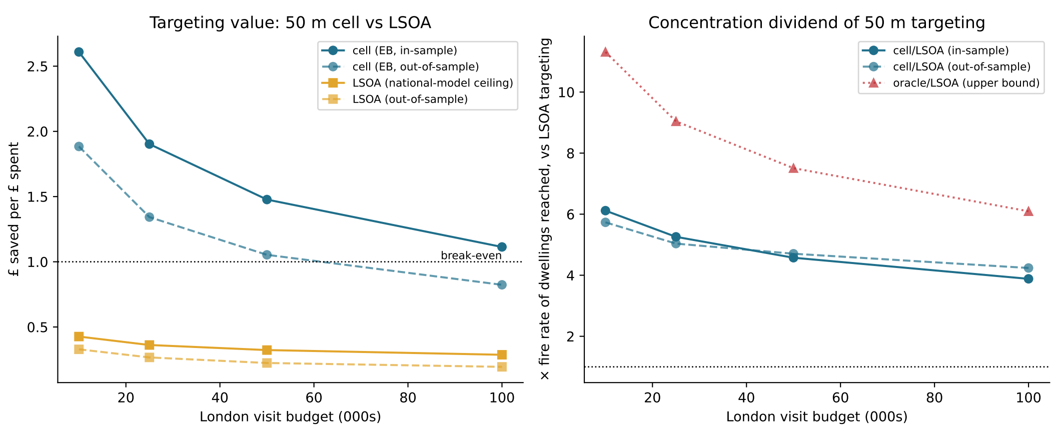

Figure 9: London pilot: out-of-sample value of sub-LSOA targeting. Cells ranked on 2009–2020 empirical-Bayes rates and scored on realised 2021–2024 rates; cell-level targeting reaches dwellings with 4.2–5.7× the fire rate of LSOA-level targeting across budgets, crossing break-even on the smoke-alarm channel alone at small budgets.

6.7 Real-building stress test: Grenfell as a model-boundary case

To probe how far the generic archetypes deviate from a real, exhaustively documented building, we ground the high-rise-flat archetype against Grenfell Tower (14 June 2017), using only public evidence from the Grenfell Tower Inquiry Phase 1 report [Grenfell Tower Inquiry, 2019]. The 72 people who died are the reason this work exists; the case study is framed in that spirit, and its purpose is to state plainly what a population-scale interior model can and cannot represent. We rebuild the tower to its real scale (24 storeys, six flats per floor, 120 residential flats, a single central stair, 297 occupants present) with the same simulation engine and no change to any parameter, and walk a one-variable-at-a-time path from the published archetype to the real building (Table 10).

The path isolates a clear finding: *geometry barely matters; regime assumptions dominate*. Placing the fire in the occupied flat — the correct picture for the exposed household under a stay-put design — raises per-occupant evacuation failure from 0.072 to 0.252 with no change in geometry. Replacing the generic 12-storey shape with Grenfell's real 24-storey geometry then moves it only a further 2% (0.252 to 0.258): doubling the storeys and adding flats per floor is nearly irrelevant to whether the origin household escapes, because the single-stair topology the archetype *already* encodes is the safety-critical feature, and its compartmentation score (1.356) essentially matches the real graph's (1.345). The catastrophe is a regime change, not a geometry effect: losing compartmentation and forcing simultaneous full evacuation takes per-occupant failure to 0.987 and expected fatalities from about 0.5 (one household) to about 293 (the whole building). The parameters that differ most between archetype and reality — the number of occupants in the evacuation problem (2, the origin household under an intact stay-put design, vs 297, the whole building once stay-put is revoked; per-flat occupancy is nearly identical at ≈2–2.5) and storeys (12 vs 24) — are *not* the parameters that move the risk; whether compartmentation holds and whether the building stays under stay-put are. The generic archetype is therefore a sound geometric proxy whose one material simplification, shared by every archetype, is that it presumes the design strategy holds.

Two honest boundaries follow. First, even in the breached case a working common alarm with early simultaneous evacuation cuts expected fatalities from about 293 to about 4 — the interior consequence of the post-Grenfell shift toward evacuation-alert systems, which the model reproduces. Second, and decisively, *our interior graph has no external facade*. The mechanism that actually destroyed Grenfell — flames running up aluminium composite material (ACM) rainscreen cladding with polyethylene cores and re-entering flats through failed windows on many floors at once — lies entirely outside a room–corridor–stair model, which is why the breached no-alarm case predicts about 293 of 297 failing when 72 in fact died: it is a worst-case envelope (no time-phased escape, a hard tenability cutoff), not a reconstruction. That blind spot is not a defect to patch inside the model; it is the boundary between population-scale interior risk ranking and the external-wall and safety-case regime — the combustible-cladding ban, Requirement B4 external-fire-spread compliance (which Grenfell failed), and the Building Safety Act's higher-risk-building regime — created precisely for the hazard our model cannot see. That division of labour is the honest conclusion. Table 9 makes the envelope explicit: each simplification in the breached scenario removes a survival mechanism that operated on the night, so every bias points the same way, and the two runs should be read as brackets — the event traversed from the intact regime toward the breached one over roughly two hours, and the actual toll reflects who was still inside when that transition completed.

Conditions (a) and (b) bracket the problem; a third condition (c) *tests* it. Feeding the model the documented Phase 1 fire timeline as an exogenous boundary condition — cladding involvement, an upward front of about one floor per minute, lobby smoke-logging from ≈01:40, the single stair carrying significant smoke from 02:00 — and simulating only the egress and tenability response, with behaviour calibrated *solely* to the recorded departure anchor ("168 of 297 out by 01:50", which the model reproduces at 176) and *never* to the death toll, the model predicts 37 expected fatalities (90% Monte Carlo [28, 46]) against the actual 72 — the right order of magnitude, under by about half (Table 10, row (c)). This tests only the egress/tenability half of the model, since the hazard field is imposed rather than derived. The miss is structured, not diffuse: a five-parameter sensitivity sweep — including the two per-floor stair-lethality knobs that dominate the fatality count, each over its stated factor-of-two range — spans [12, 99] in expectation; constrained to departure curves consistent with the observed anchor (155–185 out by 01:50), it spans [20, 73], so the true toll of 72 sits at the *upper boundary* of the anchor-respecting envelope rather than comfortably within it, and predicted fatalities rise with height as the documented toll did. The central estimate reaches 72 only by pushing every dominant knob to its lethal extreme; at plausible-central settings the tenability mechanics under-predict, which localises the missing half to factors the model deliberately does not tune away: occupant vulnerability held at the archetype's 10% impaired fraction (Grenfell's dead were disproportionately disabled, elderly and children), coarse rescue and smoke modelling, and stay-put maintained beyond survivable conditions — the human and behavioural failures the Inquiry identified. An honest, un-tuned under-prediction of this shape is more informative than a fitted hit: the geometry and tenability mechanics are sound, and the missing half lives in vulnerability and policy, not in the model's physics.

Table 9: Breached-scenario assumptions vs the recorded event (Grenfell Tower Inquiry Phase 1). Every simplification removes a survival mechanism, so the scenario is a one-sided worst-case envelope: it predicts ≈293 of 297 failing where 72 died. The intact-compartmentation run (≈0.5 expected) is the opposite envelope, assuming the stay-put design works perfectly.

Model assumption (breached run)	What happened on the night	Bias on fatalities
Compartmentation lost building-wide at ignition	Progressive failure over ∼1–2 h as the facade fire re-entered floor by floor	overstates
Unaided self-evacuation only	Dozens rescued or assisted by firefighters	overstates
Occupants move only when alerted, uniform compliance	168 of 297 had left by 01:50, many against stay-put advice, before its 02:47 revocation; deaths concentrated among those who remained longest	overstates
Stair degrades with the compartment loss	The single stair remained partially passable for hours despite smoke-logging	overstates
Hard tenability cutoff, no refuge or relocation	Some occupants moved between flats and survived long after lobbies smoke-logged	overstates

6.8 Instantiating the inspection layer on real assessments

The paper's proposed inspection-as-noisy-observation layer (Section 3) is here fitted to real data for the first time: a corpus of 1,441 published fire risk assessments over 653 Camden council estates [London Borough of Camden, 2026], linked to London Fire Brigade primary-fire histories [London Fire Brigade, 2026]. What is *established* is the apparatus, not a positive validity claim, and we state each part plainly.

The measurement model fits coherently. A joint latent-factor model over the 653 estates recovers a single continuous latent deficit state θ whose measurement model is the FRA indicators. They load coherently on it — priority-A and priority-B action counts, overall rating and the compartmentation, electrical and alarm flags all move together (only the higher-risk-building-specific

Table 10: Deviation path from the published archetype to real Grenfell geometry (rows a: night, no alarm). Per-occupant evacuation-failure probability and expected number of occupants failing to escape. Geometry moves the exposed-household risk by only 2%; the regime change (lost compartmentation + full evacuation) moves it by two orders of magnitude. Row (c) is the as-unfolded hindcast under the imposed Phase 1 timeline (per-occupant fatality; predicted vs actual 72).

Configuration	Occ.	$P(\text{fail} \mid \text{occ.})$	$\mathbb{E}[\text{n fail}]$
Generic archetype (as published)	2	0.072	0.14
Generic geometry, fire in occupied flat	2	0.252	0.50
Grenfell geometry, intact (stay put)	2	0.258	0.52
Grenfell breached, full evacuation	296	0.987	292.5
(c) As-unfolded hindcast (imposed timeline)	∼288	0.128	37 (actual 72)

external-wall flag is near zero) — so θ is a well-defined "assessed-deficiency" state. Of the between-estate log fire-rate variance attributed to θ plus the public deprivation covariate, θ carries a median 94%: the latent read off the inspection dominates the IMD covariate in explaining where estate fire rates differ.

The posterior update works numerically. For a worked single block (six flats, most-deprived IMD decile, rating Moderate, nine actions, electrical and external-wall flags), conditioning on its FRA observation through the measurement model turns a prior expected fire rate of 0.027 fires/year (95% CrI 0.016–0.048) into a posterior of 0.025 (95% CrI 0.017–0.036): the inspection both shifts and, more importantly, *narrows* the posterior (Figure 10). This is Equations (6)–(7) made fully numeric for one building — the demonstration the framework promised.

The first empirical assessor-noise bounds exist. Within-document "raised in previous report" marks identify the compound of defect persistence and assessor detection, not detection alone. Pooled recurrence fractions run 0.16–0.64 by prior-action status; among blocks whose prior actions were left incomplete (high persistence) the recurrence fraction 0.16 gives a *lower bound* on detection sensitivity q of 0.16–0.32 across stated persistence assumptions (Table 11). The compound is identified; sensitivity is only partially identified — point identification would need an actual re-inspection wave — and we claim no more.

A small true re-inspection pilot now exists: for eight blocks, an earlier assessment of the same building (median gap eleven months) was recovered from public web archives and compared item-by-item against the current one. Directly measured persistence is high — 84 of 121 wave-0 findings (69%) match a finding in the new assessment, ranging 61–83% across matching thresholds — yet *none* of the 119 current-wave findings in these blocks carries the "raised in previous report" mark, against a corpus-wide marking rate of 22%. The copy-forward marker is therefore assessor- or document-dependent rather than systematic, so the recurrence fractions above under-record true persistence and the sensitivity bounds are conservative — documentation noise of exactly the kind the measurement framework models, now observed directly, in the first repeated-measurement FRA observations we are aware of ($n = 8$ pairs; a pilot and a pipeline rehearsal for the full second wave, not a population estimate).

The validity analyses, reported honestly, are where the thesis meets the data. Do FRA outputs track a block's long-run fire propensity? Regressing pre-assessment fires (2009–23) on the later assessment, every estate-level rate ratio (RR) spans one. The one block-level association excluding one — the medium-priority (priority-B) action count, RR 1.11 [1.03, 1.21] — dies under estate-level aggregation (RR 1.00 [0.96, 1.04]), the unit robust to the fact that 54% of blocks share an estate-centroid coordinate, so we read it as a co-located-attribution artefact. The block-level compartmentation flag runs *negative* (more assessed deficiency, *fewer* historical fires), while the latent deficit state θ couples to fire rate as a null (RR 1.00 [0.69, 1.44], $P(\beta_\theta > 0) = 0.50$) — either no detectable coupling exists at this sample size, or a true positive coupling is masked by the same remediation-endogeneity mechanism the compartmentation flag exhibits — assessors flag worse blocks, which then receive remediation, cladding programmes and closer management that remove the very propensity the flag marked — *not* evidence that deficiencies are protective. The cleanest temporal test, whether prior incomplete actions predict fires in the following window, is a low-power null (140 fires over ≈3.2-year windows; all intervals span one).

The interpretation is the paper's thesis meeting real evidence. A single FRA wave cannot be read as a risk score — it is confounded by remediation endogeneity and limited by measurement noise — which is precisely why the framework treats an inspection as noisy evidence to be assimilated rather than as ground truth. What is demonstrated is the apparatus (a coherent measurement model, a numeric prior-to-posterior update, and the first bounds on assessor noise) on one borough's council stock under a retrospective validity design; external validity is unestablished, and de-confounding the structural coupling requires a second assessment wave — a Camden republication or a Freedom of Information (FOI) release of historical FRAs — the concrete data gap the programme's next stage targets.

Table 11: Partial identification of assessor detection sensitivity q. The recurrence fraction among blocks with incomplete prior actions gives a conservative *lower* bound on q under an assumed defect-persistence p; the compound persistence×detection is identified, q alone is only bounded.

Assumed persistence p	q lower bound	q lower bound (95% cons.)
1.00	0.16	0.14
0.90	0.18	0.16
0.75	0.22	0.19
0.50	0.32	0.29

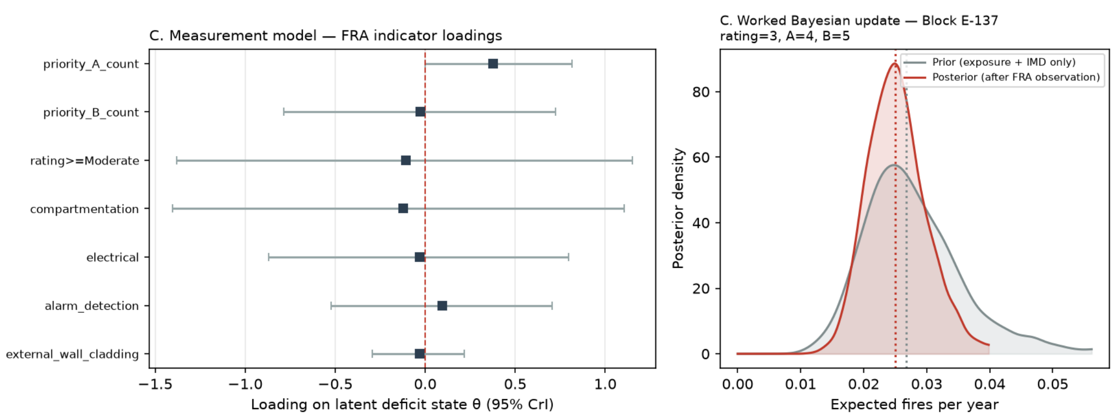

Figure 10: Worked single-block posterior update: conditioning on one estate's FRA observation shifts the expected fire-rate prior (95% CrI 0.016–0.048 fires/year) to a tighter posterior (0.017–0.036). The inspection narrows the distribution — the inspection-as-noisy-observation mechanism made numeric.

6.9 Ignition-source trends

Finally, a descriptive analysis of ignition sources over FY 2010/11–2024/25 (446,470 fires, Suffolk excluded from *every* year to hold the 42-service panel constant) fits a per-level generalised linear model (GLM) on fiscal year, reporting each source's trend in its *share* of dwelling fires (Table 12, Figure 11). The clearest structural shift is a broad, uniform decline in cooking-related ignition — food as the ignited item falls 10.8 percentage points, the single largest mover, alongside grills, hobs, ovens and chip-pan fires — consistent with safer appliance design. Against that backdrop the fastest-rising identifiable source is the battery/electrical-apparatus family: "battery charger" fires rose +425% and "apparatus — batteries, generators" +86.8%, their combined share nearly tripling (1.28% to 3.78%, share odds ratio 1.059 per year).

The most useful finding, though, is a *data-gap* one. The national taxonomy cannot isolate the hazard the sector most fears: lithium-ion e-bike and e-scooter fires have no dedicated category, and the only proxy ("batteries, generators" plus "battery charger") nets device batteries with standby generators and predates the e-mobility boom. London Fire Brigade's device-specific counts rose roughly thirtyfold over 2017–2025 while the national proxy grew only about 72% over fifteen years — a shape mismatch implying e-mobility fires are being absorbed into generic levels. We flag this to MHCLG: a dedicated device-type sub-field is needed before the national data can track this hazard. Separately, the rising "Unspecified"/"Other" buckets (+2.6 and +2.0 percentage points) are themselves a recording-drift flag — a growing "don't know" share erodes the precision of every other category, the battery proxy included.

Table 12: Biggest movers in ignition-source *share* of dwelling fires, FY 2010/11–2024/25 (share odds ratio per year from a binomial GLM). The battery family rises fastest among identifiable sources; cooking-related sources fall broadly.

Ignition source	Δ share (pp)	Share OR/yr	Trend
Battery charger	+0.44	1.125	rising (+425% count)
Apparatus: batteries, generators	+2.06	1.052	rising (+86.8%)
Combined battery proxy	+2.50	1.059	rising
Grill / toaster	−3.02	0.956	falling
Cooker incl. oven	−0.72	0.993	falling
Food (item ignited)	−10.78	—	falling (largest mover)

7 Discussion and Limitations

The framework fits the post-Grenfell regulatory direction directly. Safety cases under the Building Safety Act ask duty-holders to demonstrate that risks are understood and managed on an ongoing basis [Grenfell Tower Inquiry, 2024, UK Parliament, 2022]; a posterior risk distribution with explicit credible intervals and missing-evidence flags is a more faithful object for that purpose than a categorical score, and the expected-value-of-inspection machinery turns the same posterior into a prioritisation rule for where to spend finite inspection effort. The results support the premise: the occurrence model is calibrated on held-out years and recovers the deprivation gradient; the consequence models are calibrated out-of-time and isolate a clean detection→spread signal; and the simulator reproduces the observed spread orderings while exposing an archetype-specific mechanism (detection helps flats and houses, but not mobility-limited care homes) that a purely statistical model would miss; and the value-of-targeting study puts a conservative lower bound of a roughly ninefold return on that prioritisation over untargeted allocation. Taken together they move FRA

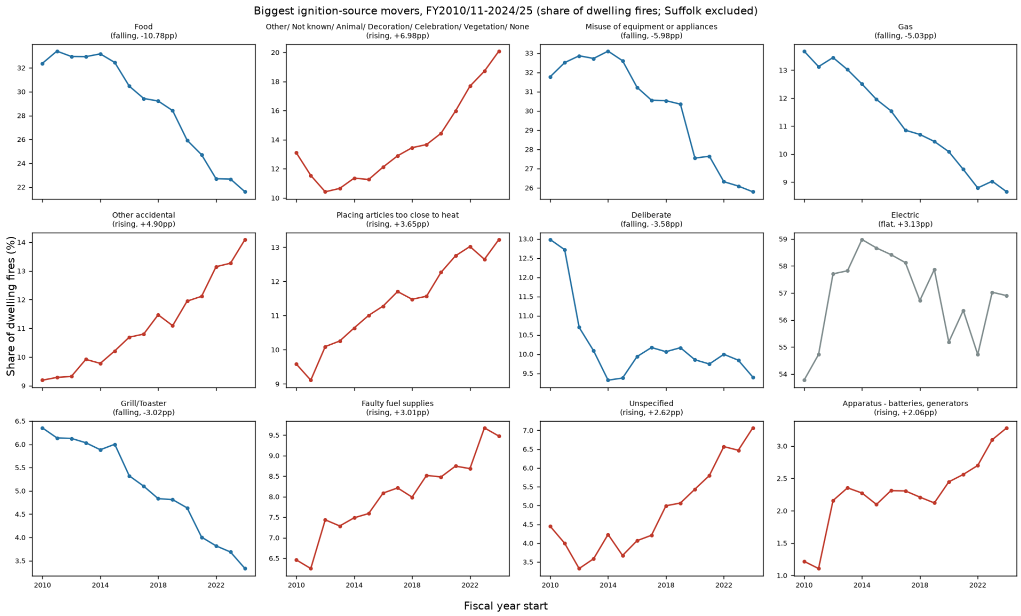

Figure 11: Share trajectories of the twelve biggest-moving ignition sources, coloured by trend classification. Cooking sources decline broadly; the battery/apparatus family is the clearest riser after the generic "Unspecified"/"Other" buckets.

from a static compliance checklist [British Standards Institution, 2020a,b] toward an updateable, evidence-based risk picture — with the fully sequential, inspection-assimilating version specified in Section 3 as the programme's next stage.

The real-building case study sharpens what the archetypes are for. Grounding the high-rise-flat archetype against Grenfell Tower shows that its scale parameters — storeys and occupant load — matter far less to modelled risk than its regime assumptions: doubling the storeys shifts the exposed-household failure probability by only 2%, whereas losing compartmentation and abandoning stay-put shifts expected fatalities by two orders of magnitude. The archetype is a sound geometric proxy precisely because it already encodes the safety-critical single-stair topology; its unavoidable simplification is that, like every archetype, it presumes the design strategy holds. The same study marks the framework's outer boundary: an interior room–corridor–stair graph cannot represent external cladding-driven spread, the mechanism that drove Grenfell, and must defer to the external-wall and safety-case regime of the Building Safety Act for that hazard. A real-geometry v2 pipeline — replacing archetype graphs with per-building layouts drawn from the downloadable building-specification sources we catalogue (`docs/building_specs_sources.md`) — is the natural next step, but the external-wall hazard will remain a regulated, per-building duty rather than something a population-scale model should attempt.

Several limitations bound these claims, and we have kept them visible rather than smoothing them over. This is an England-only residential MVP built on archetype graphs, not real building geometry; the simulator is a transparent stochastic caricature, not fire-physics CFD, and its multi-storey houses over-predict spread beyond the floor of origin because of an idealised open internal stair; the care-home archetype's building-level evacuation-failure rate overstates the observed communal-living casualty rate by roughly 1.4× per occupant — and far more (∼4.8×) at the building-level any-failure measure — reflecting its no-staff-assist, stay-put-free full-evacuation assumption. Both headline posteriors are exact full-data objects — the consequence models by

NUTS on all 451,172 records, the occurrence model by PSIS-corrected Laplace importance sampling on all 202,530 training rows ($\hat{k} = 0.33$; Section 3) after full-data NUTS failed under three parameterisations — each corroborated by subsample NUTS, ADVI and MLE cross-checks; the occurrence model's linear time trend under-predicts the steeper 2024/25 decline. The consequence models are covariate-adjusted associations, not causal effects: the casualty–alarm odds ratio falls below one purely through occupancy confounding, which is precisely why the model carries a latent-state layer, but that layer de-confounds by assumption and structure rather than by identification, and this paper reports the adjusted associations that feed it rather than a validated causal estimate. The FRS response-time covariate carries little trustworthy independent signal: it is ecological (identified at FRS×year resolution and entangled with the secular trend), and its effect on spread is not even robust in sign — below one on the subsample, above one at full data — so we retain it only as a control and report no directional response-time effect. The exact full-data fit exposes a further honest gap: it undercovers per-service held-out counts (an FRS-overdispersion a single random intercept misses), so per-service predictive intervals should be read as anti-conservative and a random-slope extension is future work. Data linkage is imperfect — record-level fires resolve only to FRS area, not LSOA; covariates mix vintages (IMD 2025, Census 2021, EPC 2012–2026); and EPC attributes enter only as LSOA aggregates, consistent with the address-level redistribution restriction of Section 4. Finally, fire is a rare event, so building-level validation is not possible from open data: our calibration is at the aggregate (LSOA-year and FRS-area) level, and the worked building-level decomposition of Section 6.5 is illustrative of the composition, not a validated per-building prediction.

Equity of targeted prevention. Because fire risk is strongly deprivation-linked, any risk-targeted allocation concentrates visits in deprived areas. We regard this as the intended behaviour of a prevention programme — resources flow to where expected harm is greatest — but it carries obligations the framework should make explicit rather than bury in a score: targeting criteria must be auditable (a virtue of interpretable rate ratios over black-box scores), the uncertainty attached to any area's ranking should be visible to decision-makers, and allocations should be monitored for unintended effects, such as areas of genuinely high but poorly evidenced risk (wide posteriors) being systematically deprioritised relative to areas that are merely well measured. The expected-benefit policy of Section 6.6 partially addresses the latter by valuing uncertainty reduction, but a full fairness analysis of risk-targeted fire prevention — who gains, who is overlooked, and under which valuation of harm — is future work we consider necessary before operational deployment.

The physics programme. The simulator's mechanism-informed status is a staging point, not an endpoint. A companion work builds the corresponding physics engine (NIST CFAST zone models over the same archetypes, with era-derived boundary materials and design fires keyed to the national ignition-source distribution) and calibrates its parameters against the observed English spread margins with a model-discrepancy term. Its headline is a caution this paper's framing anticipates: the calibrated physics reproduces the aggregate spread surface, but aggregate margins identify only one physical parameter (effective door-openness) — the remainder require *timed*, scenario-resolved calibration targets, which is a further argument for the record-level data access discussed above.

Secondary findings with policy implications. Two supporting analyses sharpen the operational picture. The no-alarm spread penalty is not uniform across households — it is roughly twice the pooled value for lone pensioners (Section 6.3) — giving the targeting layer a concrete, data-backed reason to up-weight areas rich in that group beyond what their fire rate alone implies, precisely where consequence decouples from occurrence, while the night penalty is a whole-stock effect that warrants no dwelling-specific multiplier. And the ignition-source analysis surfaces a data-infrastructure gap: England's central incident taxonomy cannot isolate lithium-ion e-bike and e-scooter fires, the hazard the sector is most anxious about, so the most actionable recommendation there is not a model change but a taxonomy one (Section 6.9).

The inspection layer, meeting data. The sharpest test of the paper's central claim — that an FRA is a noisy observation, not a risk score — is the first instantiation of the inspection layer on real assessments (Section 6.8): the apparatus runs end to end on 1,441 published assessments and a single inspection demonstrably narrows a block's fire-rate posterior, yet on one cross-sectional wave the latent's coupling to fire rate is a remediation-confounded null, so a snapshot FRA is *not* a validated proxy for persistent fire propensity. That is exactly the situation the framework is designed for — inspections assimilated as noisy evidence under explicit uncertainty — and it names the missing ingredient: a second assessment wave to turn the retrospective check into a prospective test and point-identify the assessor-noise parameters.

8 Conclusion

We have recast UK residential fire risk assessment as Bayesian inference: risk is a posterior distribution over ignition, spread, evacuation failure, and consequence, updateable as evidence arrives, rather than a categorical score. The instantiated components stand on their own evidence: a small-area occurrence model whose 50% and 90% intervals cover 50.5% and 90.2% of held-out outcomes across all 33,755 English LSOAs, with calibration holding within deprivation, geography, and building-form subgroups; consequence models that separate predictive association from intervention effect and expose, rather than inherit, the confounds of observational fire data; a mechanism-informed simulator whose orderings match the national record and whose boundary a real-building stress test locates precisely; and a decision layer in which the same posteriors allocate a fixed prevention budget to roughly nine times the value of untargeted allocation — above break-even outright at small, concentrated budgets. The proposed latent-state layer is now partially real: its measurement model is fitted on 1,441 published fire risk assessments, a single inspection demonstrably narrows a building's risk posterior, and the first empirical bounds on assessor detection sensitivity exist, with a recovered two-wave pilot showing directly that findings persist while the documentation trail under-records them. The novelty remains the integration — no prior work couples these elements in one auditable pipeline — but the integration now carries evidence at every joint. What stands between this framework and live, building-resolved safety intelligence is data, not method: repeated assessment waves to point-identify assessor noise and intervention effectiveness, fine incident geography to nationalise the demonstrated sub-area targeting gains, and timed outcomes to push physics-calibrated consequences past their identifiability ceiling — each a concrete, named acquisition rather than an open research question.

A Reproducibility

Every numbered figure and table in this paper traces to a results artifact under `results/`, which traces to a module under `src/` that reads the project database. The end-to-end map, environment (`requirements.txt`, frozen versions; Python 3.12), and the one-command database rebuild (`python -m src.build_db -with-epc -with-lfb`, or attach the shipped DuckDB files) are documented in

REPRODUCE.md at the repository root. Priors are exactly as stated in the model sections (Sections 3–5); all sampler settings (draws, warmup, chains, subsample sizes) live in the cited modules and are echoed into each `results/*/summary.md`. All Monte Carlo and MCMC runs use fixed seeds, listed below. Tables 13 and 14 give the per-figure and per-table provenance; where a figure was regenerated from a saved CSV rather than a module entry point, that is stated.

Table 13: Reproducing each numbered *figure*. Commands are run from the repository root inside the virtual environment; seeds are the module-level `SEED` constants.

Figure	Producing module / function	Input artifacts	Seed(s) & config
2	`src.models.evaluate_occurrence` (`fig_calibration`); via `run_occurrence eval`	occurrence posterior (`results/occurrence/`); held-out FY2023/24	`SEED=20240705`
3	`src.models.evaluate_occurrence` (`decile_gradient_fig`)	occurrence posterior; IMD deciles	`SEED=20240705`
5	`src.models.consequence` (`fig_reliability`); regenerated from the out-of-time prediction CSV, not a standalone entry point	`results/consequence/` spread predictions	`SEED=20260705`
6	`src.models.interactions` (`fig_ame_heatmap`)	`dwelling_fires` (DB); alarm×occupancy design	`SEED=20260705`; NUTS 4 × (800+1000)
7	`src.sim.make_report` (`fig_failure_by_archetype`)	`results/sim/mc_summary.csv`	`BASE_SEED=20260705`
1	`python -m src.sim.morris_sweep`	`results/sim/morris_ee.csv`, `morris_ranking.csv`	`BASE_SEED=20260713`; 20 trajectories, 6 levels, 4,000 runs/cell
8	`src.models.decision` (`fig_sensitivity`)	`results/` occurrence + consequence engines	`SEED=20260705`
9	`src.models.london_pilot` (`fig_targeting`)	London 50 m cell panel (`results/london_pilot/`)	`SEED=20260706`
10	`src.models.latent_state` (`figC_measurement_worked_update`)	Camden FRA panel (`data/processed/fra/`)	`SEED=20260706`
11	`src.models.ignition_trends` (`small_multiples_movers`)	`dwelling_fires` ignition-source shares	deterministic (no seed)

References

Vytenis Babrauskas, Jay M. Fleming, and B. Don Russell. Rset/aset, a flawed concept for fire safety assessment. *Fire and Materials*, 34(7):341–355, 2010. doi: 10.1002/fam.1025.

British Standards Institution. PAS 79-1:2020 fire risk assessment. premises other than housing – code of practice. Technical report, BSI, London, 2020a.

British Standards Institution. PAS 79-2:2020 fire risk assessment. housing – code of practice. Technical report, BSI, London, 2020b.

British Standards Institution. BS 8674:2025 built environment. framework for competence of individual fire risk assessors – specification. Technical report, BSI, London, 2025.

Francesca Campolongo, Jessica Cariboni, and Andrea Saltelli. An effective screening design for sensitivity analysis of large models. *Environmental Modelling & Software*, 22(10):1509–1518, 2007.

Jinyue Chen, Long Ding, and Jie Ji. A combined method to build bayesian network for fire risk assessment of historical buildings. *Fire Technology*, 59:3525–3552, 2023. doi: 10.1007/s10694-023-01475-8.

Hao Cheng and George V. Hadjisophocleous. The modeling of fire spread in buildings by bayesian network. *Fire Safety Journal*, 44(6):901–908, 2009. doi: 10.1016/j.firesaf.2009.05.005.

He-in Cheong, Zhiyu Wu, Arnab Majumdar, and Washington Yotto Ochieng. One-way coupling of fire and egress modeling for realistic evaluation of evacuation process. *Transportation Research Record*, 2675(12): 321–334, 2021. doi: 10.1177/03611981211032636.

Table 14: Reproducing each numbered *table*. Numbers are summarised into the named `results/*/summary.md` by the producing module.

Table	Producing module / command	Input artifacts	Seed(s) & config
1	none — hand-built from `data/sources/*.md`	public source catalogue	n/a (see `REPRODUCE.md` §1)
2	`python -m src.models.run_occurrence fit`	occurrence posterior (`results/occurrence/summary.md`)	`SEED=20240705`; exact full-data posterior via `python -m src.models.full_is` (`SEED=20260714`; unimodality probe `full_is multistart` → `fullis_multistart.csv`) / `full_exact` (20260713)
3	`python -m src.models.spatial_ablation fit` / `eval`	`results/occurrence/` (`ladder.log`, `_v2`)	`SEED=20240706`
4	`python -m src.models.consequence`	`results/consequence/summary.md`	`SEED=20260705`; NUTS $4 \times (800{+}1000)$
5	`python -m src.models.consequence`	held-out proper scores (`results/consequence/`)	`SEED=20260705`; NUTS $4 \times (800{+}1000)$
6	`python -m src.sim.run_experiments -runs 10000`	`results/sim/mc_summary.csv`	`BASE_SEED=20260705` (+101 per cell), N=10,000
7	`python -m src.models.decision`	`results/decision/summary.md`	`SEED=20260705`
8	`python -m src.models.decision` (`fig_budget_sweep` companion)	`results/decision/`	`SEED=20260705`
9	`python -m src.sim.case_study`	`results/case_study/summary.md`	`BASE_SEED=20260705`
10	`python -m src.sim.case_study` & `src.sim.hindcast`	`results/case_study/`	`BASE_SEED=20260705`
11	`python src/models/latent_state.py run`	`results/latent_state/summary.md`	`SEED=20260706`
12	`python -m src.models.ignition_trends`	`results/ignition_trends/summary.md`	deterministic (no seed)

Louis Anthony (Tony) Cox. What's wrong with risk matrices? *Risk Analysis*, 28(2):497–512, 2008. doi: 10.1111/j.1539-6924.2008.01030.x.

Zhen Dai, Ying Zhang, and Xinyan Huang. Fire prediction and risk identification with interpretable machine learning. *Journal of Forecasting*, 2025. doi: 10.1002/for.3266.

Jamiu A. Dauda, Muhammad A. Yamusa, Adekunle S. Oyegoke, Saheed O. Ajayi, and Abdullahi B. Saka. A systematic review of building fire safety practices in the UK: Approaches, challenges and recommendations. *Fire Technology*, 61, 2025. doi: 10.1007/s10694-025-01801-2.

Wouter Jan de Klerk, Timo Schweckendiek, Frank den Heijer, and Matthijs Kok. Value of information of structural health monitoring in asset management of flood defences. *Infrastructures*, 4(3):56, 2019. doi: 10.3390/infrastructures4030056.

Paul Embrechts, Claudia Klüppelberg, and Thomas Mikosch. *Modelling Extremal Events for Insurance and Finance*. Springer, Berlin, 1997. doi: 10.1007/978-3-642-33483-2.

Julen Erzibengoa, Meritxell Gómez-Omella, and Izaro Goienetxea. IberFire – a detailed creation of a spatio-temporal dataset for wildfire risk assessment in Spain. *arXiv preprint arXiv:2505.00837*, 2025. doi: 10.48550/arXiv.2505.00837.

Hector Diego Estrada-Lugo, Marco De Angelis, and Edoardo Patelli. Probabilistic risk assessment of fire occurrence in residential buildings: Application to the Grenfell tower. In *Proceedings of the 13th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP13)*, Seoul, South Korea, 2019.

Stanley W. Gilbert and David T. Butry. Estimating smoke alarm effectiveness in homes. *Fire Technology*, 57:1497–1516, 2021. doi: 10.1007/s10694-020-01072-z.

Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society: Series B*, 69(2):243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.

Grenfell Tower Inquiry. Grenfell tower inquiry: Phase 1 report. Report of the Public Inquiry, Chair Sir Martin Moore-Bick, 4 volumes, HC 49-I to 49-IV, 2019. https://www.grenfelltowerinquiry.org.uk/phase-1-report.

Grenfell Tower Inquiry. Grenfell tower inquiry: Phase 2 report. Report of the Public Inquiry, Chair Sir Martin Moore-Bick, 7 volumes, HC 19-I to 19-VII, 2024. https://www.grenfelltowerinquiry.org.uk.

Judith Hackitt. Building a safer future: Independent review of building regulations and fire safety — final report. Technical report, HM Government (Ministry of Housing, Communities and Local Government), London, 2018. URL https://www.gov.uk/government/publications/independent-review-of-building-regulations-and-fire-safety-final-report. Cm 9607.

Daniela Hanea and Ben Ale. Risk of human fatality in building fires: A decision tool using bayesian networks. *Fire Safety Journal*, 44(5):704–710, 2009. doi: 10.1016/j.firesaf.2009.01.006.

Edwin Hidayat, Ben Seligmann, and Jurij Karlovšek. Improving tunnel fire safety: A bayesian network model for risk analysis and a novel risk acceptability range for risk evaluation. *Australian Journal of Civil Engineering*, 2025. doi: 10.1080/14488353.2025.2465102.

Home Office. The economic and social cost of fire. UK Government statistical publication and data tables, 2023. https://www.gov.uk/government/publications/economic-and-social-cost-of-fire.

Kieran Kalair and Colm Connaughton. Dynamic and interpretable hazard-based models of traffic incident durations. *Frontiers in Future Transportation*, 2:669015, 2021. doi: 10.3389/ffutr.2021.669015.

Brian Y. Lattimer, Jonathan L. Hodges, and Amanda M. Lattimer. Using machine learning in physics-based simulation of fire. *Fire Safety Journal*, 114:102991, 2020. doi: 10.1016/j.firesaf.2020.102991.

Chih-Chieh Lee, Ya-Wen Hsu, and Chen-Yu Wang. Machine learning-based forecasting of urban fire impact in city environments. *Science Progress*, 2025. doi: 10.1177/00368504251406566.

Fangyuan Li, Yijing Li, and Luke Edward Rogerson. London blue light collaboration evaluation: A comparative analysis of spatio-temporal patterns on emergency services by london ambulance service and london fire brigade, 2025. URL https://arxiv.org/abs/2506.06011.

Xuewei Li, Jingfeng Yuan, and Limao Zhang. Risk assessment of subway station fire by using a bayesian network-based scenario evolution model. *Journal of Civil Engineering and Management*, 30, 2024. doi: 10.3846/jcem.2024.20846.

Yijing Li, Aliyu Musah, Andrew Thomas, and Patty Kostkova. An ecological study exploring the geospatial associations between socioeconomic deprivation and fire-related dwelling casualties in england (2010–2019). *Applied Geography*, 145:102718, 2022. doi: 10.1016/j.apgeog.2022.102718.

London Borough of Camden. Fire risk assessments — housing (open data). Camden Open Data portal, 2026. https://opendata.camden.gov.uk.

London Fire Brigade. London fire brigade incident records. London Datastore, Greater London Authority, 2026. https://data.london.gov.uk/dataset/london-fire-brigade-incident-records-em8xy.

Ying Lu, Xuehui Fan, and Yong Zhang. Machine learning models using shapley additive explanation for fire risk assessment mode and effects analysis of stadiums. *Sensors*, 23(4):2151, 2023. doi: 10.3390/s23042151.

Jesus Luque and Daniel Straub. Risk-based optimal inspection strategies for structural systems using dynamic bayesian networks. *Structural Safety*, 76:68–80, 2019. doi: 10.1016/j.strusafe.2018.08.002.

Ying C. MacNab. Hierarchical bayesian spatial modelling of small-area rates of non-rare disease. *Statistics in Medicine*, 22(10):1761–1773, 2003. doi: 10.1002/sim.1463.

D. B. Matellini, A. D. Wall, I. D. Jenkinson, J. Wang, and R. Pritchard. A three-part Bayesian network for modeling dwelling fires and their impact upon people and property. *Risk Analysis*, 38(10):2087–2104, 2018. doi: 10.1111/risa.13113.

D. Bruno Matellini, Alan D. Wall, Ian D. Jenkinson, Jin Wang, and Rob Pritchard. Modelling dwelling fire development and occupancy escape using bayesian network. *Reliability Engineering & System Safety*, 114:75–91, 2013. doi: 10.1016/j.ress.2013.01.001.

Alexander J. McNeil. Estimating the tails of loss severity distributions using extreme value theory. *ASTIN Bulletin*, 27(1):117–137, 1997. doi: 10.2143/ast.27.1.563210.

Ministry of Housing, Communities and Local Government. Energy performance of buildings data: England and wales. Open data register, GOV.UK, 2025. https://epc.opendatacommunities.org.

Ministry of Housing, Communities and Local Government. Fire and rescue incident statistics, england, year ending december 2025. Accredited Official Statistics, GOV.UK, 2026. https://www.gov.uk/government/statistics/fire-and-rescue-incident-statistics-year-ending-december-2025.

Max D. Morris. Factorial sampling plans for preliminary computational experiments. *Technometrics*, 33(2):161–174, 1991.

M. Z. Naser. Mechanistically informed machine learning and artificial intelligence in fire engineering and sciences. *Fire Technology*, 57:2741–2784, 2021. doi: 10.1007/s10694-020-01069-8.

Kate T. Q. Nguyen, Yousef Abu-Zidan, and Guomin Zhang. Machine learning-based surrogate model for calibrating fire source properties in fds models of façade fire tests. *Fire Safety Journal*, 130:103591, 2022. doi: 10.1016/j.firesaf.2022.103591.

Office for National Statistics. Census 2021: Accommodation type, tenure and household characteristics, england and wales. ONS, GOV.UK, 2022. https://www.ons.gov.uk/census.

Judea Pearl. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, San Francisco, CA, 1988.

Andrew Perry. Fire safety in high-rise residential buildings. In *The Building Safety Act 2022: Questions and Answers*. Emerald Publishing, 2025. doi: 10.1108/978-1-83662-464-620251013.

Ganapathy Ramachandran. Extreme value theory and large fire losses. *ASTIN Bulletin*, 7(3):293–310, 1974. doi: 10.1017/s0515036100006115.

David Rasbash, Ganapathy Ramachandran, Baldev Kandola, John Watts, and Margaret Law. *Evaluation of Fire Safety*. John Wiley & Sons, 2004. doi: 10.1002/0470020083.

Benjamin Schröder, Lukas Arnold, and Armin Seyfried. A map representation of the aset-rset concept. *Fire Safety Journal*, 115:103154, 2020. doi: 10.1016/j.firesaf.2020.103154.

Daniel Straub and Michael H. Faber. Risk based inspection planning for structural systems. *Structural Safety*, 27(4):335–355, 2005. doi: 10.1016/j.strusafe.2005.04.001.

Lei Su, Chongwen Wei, Fan Yang, Lei Zhang, Yu Shen, Fan Zhang, and Zhichun Yang. Electrical fire dynamic risk assessment for high-rise buildings based on variable fuzzy set theory and Bayesian network. *Mathematical Problems in Engineering*, 2023:9068958, 2023. doi: 10.1155/2023/9068958.

Samson Tan, Darryl Weinert, Paul Joseph, and Khalid Moinuddin. Sensitivity and uncertainty analyses of human and organizational risks in fire safety systems for high-rise residential buildings with probabilistic T-H-O-Risk methodology. *Applied Sciences*, 11(6):2590, 2021. doi: 10.3390/app11062590.

Marion Tancogne-Dejean and Patrick Laclémence. Fire risk perception and building evacuation by vulnerable persons: Points of concern. *Fire Safety Journal*, 80:9–19, 2016. doi: 10.1016/j.firesaf.2015.11.009.

Arash Teymori Gharah Tapeh and M. Z. Naser. Artificial intelligence, machine learning, and deep learning in structural engineering: A scientometrics review of trends and best practices. *Archives of Computational Methods in Engineering*, 30:115–159, 2022. doi: 10.1007/s11831-022-09793-w.

Michael Taylor, Jane Fielding, and Sarah Dean. Behavioural trends and accidental dwelling fires. *Safety and Reliability*, 2026. doi: 10.1080/09617353.2026.2633873.

Philip Thomas, Reidar B. Bratvold, and J. Eric Bickel. The risk of using risk matrices. *SPE Economics & Management*, 6(2):56–66, 2014. doi: 10.2118/166269-PA.

UK Parliament. The regulatory reform (fire safety) order 2005. SI 2005/1541, 2005. https://www.legislation.gov.uk/uksi/2005/1541.

UK Parliament. Fire safety act 2021. c. 24, 2021. https://www.legislation.gov.uk/ukpga/2021/24.

UK Parliament. Building safety act 2022. c. 30, 2022. https://www.legislation.gov.uk/ukpga/2022/30.

Aki Vehtari, Daniel Simpson, Andrew Gelman, Yuling Yao, and Jonah Gabry. Pareto smoothed importance sampling. *Journal of Machine Learning Research*, 25(72):1–58, 2024.

Eline Vereecken, Wouter Botte, Geert Lombaert, and Robby Caspeele. Bayesian decision analysis for the optimization of inspection and repair of spatially degrading concrete structures. *Engineering Structures*, 220: 111028, 2020. doi: 10.1016/j.engstruct.2020.111028.

Konstantinos Vogiatzoglou, Costas Papadimitriou, Vasilis Bontozoglou, and Konstantinos Ampountolas. Physics-informed neural networks for parameter learning of wildfire spreading. *Computer Methods in Applied Mechanics and Engineering*, 434:117545, 2025. doi: 10.1016/j.cma.2024.117545.

Jing Wang, Yang Chen, and Long Chen. Simulation of fire smoke diffusion and personnel evacuation in large-scale complex medical buildings. *Buildings*, 15 (8):1329, 2025. doi: 10.3390/buildings15081329.

Kai Wang, Long Hou, and Bo Li. A novel fire risk assessment approach for large-scale commercial and high-rise buildings. *International Journal of Environmental Research and Public Health*, 18(13):7187, 2021a. doi: 10.3390/ijerph18137187.

Zilong Wang, Wai Cheong Tam, and Yaping Jia. P-flash — a machine learning-based model for flashover prediction using recovered temperature data. *Fire Safety Journal*, 122:103341, 2021b. doi: 10.1016/j.firesaf.2021.103341.

Sara Waring, Jane Fielding, and Michael Thomas. Examining the effectiveness and economic benefits of home fire safety visits. *Journal of Risk Research*, 2024. doi: 10.1080/13669877.2024.2447261.

Xingjian Wu, Zhaoru Li, and Jian Liu. Developing surrogate models for the early-stage design of residential blocks using graph neural networks. *Building Simulation*, 2025. doi: 10.1007/s12273-025-1237-7.

Jing Xin and Chongfu Huang. Fire risk assessment of residential buildings based on fire statistics from china. *Fire Technology*, 49:1147–1162, 2013. doi: 10.1007/s10694-013-0327-8.

Wei Xu, Yan Jiang, and Ming Zhou. Modeling and prediction fire evolution in building engineering: A knowledge graph and machine learning approach. *Journal of Asian Architecture and Building Engineering*, 2025. doi: 10.1080/13467581.2025.2603751.

Ramin Yarmohammadian, Florian Put, and Ruben Van Coile. Physics-informed surrogate modelling in fire safety engineering: A systematic review. *Applied Sciences*, 15(15):8740, 2025. doi: 10.3390/app15158740.

Mehmet Akif Yıldız. Machine learning for fire safety in the built environment: A bibliometric insight into research trends and key methods. *Buildings*, 15(14): 2465, 2025. doi: 10.3390/buildings15142465.

Boyang Yu, Sujit Ghosh, and Tobias Finn. A probabilistic approach to wildfire spread prediction using a denoising diffusion surrogate model. *Geoscientific Model Development*, 19:1027–1050, 2026. doi: 10.5194/gmd-19-1027-2026.

David Yung. *Principles of Fire Risk Assessment in Buildings*. John Wiley & Sons, 2008. doi: 10.1002/9780470714065.

Yu Zeng and Xinyan Huang. Smart building fire safety design driven by artificial intelligence. In *Interpretable Machine Learning for the Analysis, Design, Assessment, and Informed Decision Making for Civil Infrastructure*. Elsevier, 2024. doi: 10.1016/b978-0-12-824073-1.00011-3.

Yu Zeng, Xiqiang Zhang, and Longxing Su. Artificial intelligence tool for fire safety design (ifetool): Demonstration in large open spaces. *Case Studies in Thermal Engineering*, 36:102483, 2022. doi: 10.1016/j.csite.2022.102483.

Yu Zeng, Haoran Zheng, and Xiqiang Zhang. Ai-powered fire engineering design and smoke flow analysis for complex-shaped buildings. *Journal of Computational Design and Engineering*, 11(3):359–375, 2024. doi: 10.1093/jcde/qwae053.

Wei-Heng Zhang, Da-Gang Lu, Jianjun Qin, Sebastian Thöns, and Michael H. Faber. Value of information analysis in civil and infrastructure engineering: A review. *Journal of Infrastructure Preservation and Resilience*, 2:16, 2021. doi: 10.1186/s43065-021-00027-0.